

Bluetooth® Channel Sounding

A technical overview

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1. Introduction

Bluetooth® Low Energy (LE) is well-known the world over for providing users with wireless data transfer and audio capabilities. The technology is in our pockets thanks to its incorporation within our ever-smarter phones. It's on our wrists, in smart watches, and in fitness trackers. It's in our cars, allowing hands-free control and communication. And it's in our ears, enabling the streaming of high-quality audio from personal music devices and from broadcast sources via the new Bluetooth LE Audio capability, [Auracast™ broadcast audio](#).

But for many years, Bluetooth LE has also been establishing itself as a pervasive and [reliable](#) technology upon which to build device positioning applications. Bluetooth LE can be used to detect and report the presence of another device in the immediate environment, to estimate the distance between devices, and to calculate the direction in which another device can be found. These device positioning capabilities have been used to enable a wide range of applications, including digital key, asset tracking, Find My, and indoor navigation.

Bluetooth technology has continuously improved over its 25-year history. It has taken a positive evolutionary path that has yielded a series of remarkable new features and improvements to the results that products can achieve with it.

The update to the Bluetooth Core Specification adds a new feature called Bluetooth® Channel Sounding which enables secure fine ranging between two Bluetooth devices and is the topic of this paper. This paper is not intended to replace or be a substitute for the Bluetooth Core Specification.

2. Background

2.1 Device Positioning and Bluetooth LE

Bluetooth LE was first specified in 2010. From that point on, a number of key events in the evolution of Bluetooth LE as a technology for location services can be identified.

2.1.1 Bluetooth® Find Me

The same year that Bluetooth LE was first included in the Bluetooth Core Specification, the first formal location-related Bluetooth LE profile specification was released. This was the Find Me Profile.

The Find Me Profile defines a standard approach to personal item finding, also known as Find My. One device assumes the role of Find Me Locator. This is usually a smart phone. Other devices, which the user perhaps has a history of misplacing (keys with a Bluetooth key fob are a favorite), are paired with the Find Me Locator device and each assume the role of Find Me Target.

The Target device implements a GATT¹ service called the Immediate Alert Service.

When the user needs to be assisted in finding a misplaced device, they run an application on their smartphone. The application executes a device discovery procedure by scanning for advertising packets being broadcast by the missing device. Having discovered the Target device, the Locator connects to it. The application's user interface (UI) indicates that this has been done. The user typically then presses a button on the UI. This causes the application to write to the Alert Level characteristic which belongs to the Immediate Alert service. The Target device responds to the change of Alert Level value in some suitable way, perhaps emitting a loud beeping sound, flashing LEDs on and off or both. At this point, the user realizes their keys were in their jacket pocket all along, have fallen down the back of the sofa, or they are somewhere less predictable. Either way, Bluetooth technology saves the day and the user, and their lost item is reunited.

Bluetooth® Find Me is an example of a presence application. Bluetooth LE is used to determine that the lost device is nearby but does not provide an indication of its direction or distance from the Locator.

2.1.2 Beacons and First-Generation Distance Estimation

Bluetooth beacons leverage the advertising capability of Bluetooth LE. Advertising involves broadcasting small packets of data which any device in range can receive by scanning.

In 2013, Apple released the specification for the iBeacon format. This became a popular format for the content of the payloads that a beacon device would broadcast. The data in an iBeacon message includes a field called TX Power which contains a value representing the signal strength that can be expected if measured at a distance of one meter from the beacon. It was the presence of the TX Power field in iBeacon messages as well as in other comparable beacon data formats such as Google's Eddystone, which heralded the arrival of the first generation of Bluetooth LE distance estimation.

This early version of Bluetooth distance estimation involved the use of two data values and some simple physics and works like this:

- The TX Power field in beacon messages provides a reference power level at a known distance such as one meter.
- The Received Signal Strength Indicator (RSSI) associated with each received beacon message quantifies the signal strength at the receiving device.
- Physics defines a theoretical relationship between the rate at which the signal strength diminishes the further away from the transmitter that it is measured. Specifically, the signal strength at a receiver is inversely proportional to the square of its distance from the transmitter.
- The reduction of measured signal strength as we move further away from the transmitter is called Path Loss or Attenuation. In the case of iBeacon transmissions, $\text{path loss} = \text{TX Power} - \text{RSSI}$.
- Thus, knowing a reference power level at a fixed distance, the measured RSSI of a received beacon transmission and the inverse square relationship between distance and path loss, attenuation can be used to estimate the distance between the beacon and the receiver.

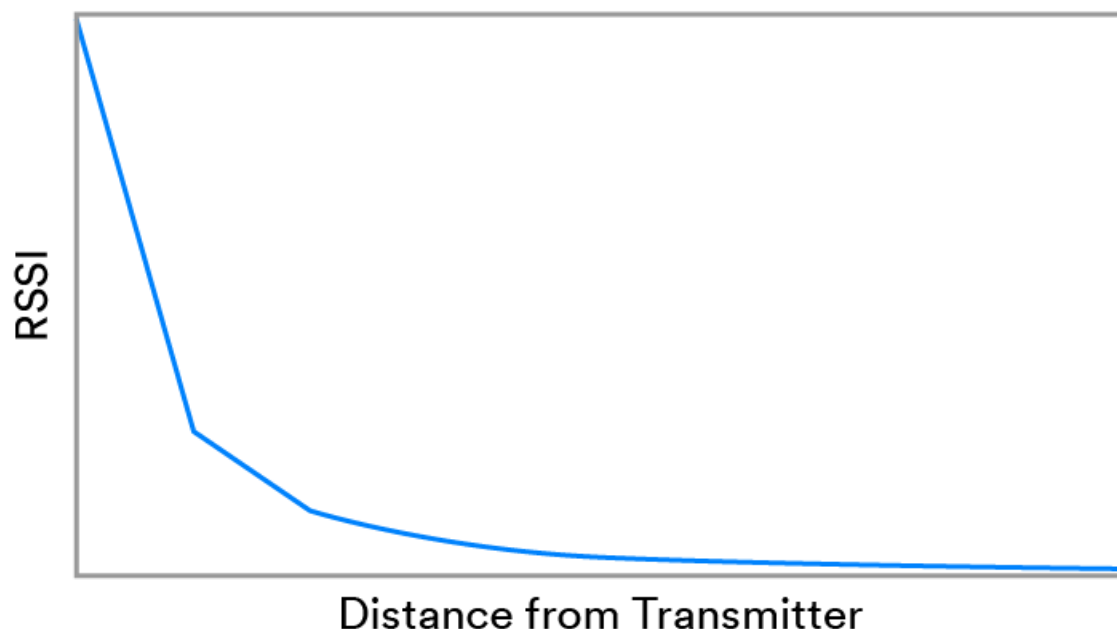


Figure 1 - Path loss and distance

Being able to estimate distance like this was quite a breakthrough, and beacons have become popular in all sorts of applications, such as retail, travel, and museums.

While beacons were an excellent fit for some requirements, distance measurements based on RSSI and path loss are not sufficiently accurate for other applications. Lacking an indication of the direction of the

transmitter is also a limitation when location data was needed rather than just proximity. Furthermore, the various proprietary beacon types such as iBeacon do not incorporate any explicit security safeguards.

2.1.3 Bluetooth® Direction Finding With AoA and AoD

In 2019, version 5.1 of the Bluetooth Core Specification included a major new feature, Bluetooth Direction Finding.

The Bluetooth Direction Finding feature enables applications to accurately calculate the direction of a received signal using phase measurements made by the Bluetooth LE controller. Two methods are defined.

In the Angle of Arrival (AoA) method, the receiving device has an antenna array and measurements of the received signal taken at different antennas exhibit phase differences due to the slightly different distances of each of its antennas to the single antenna in the transmitting device.

In the Angle of Departure (AoD) method, the transmitting device has an antenna array. The receiving device has a single antenna but possesses details of the antenna array in the remote, transmitting device. This enables it to make similar calculations from phase measurements made at its single antenna.

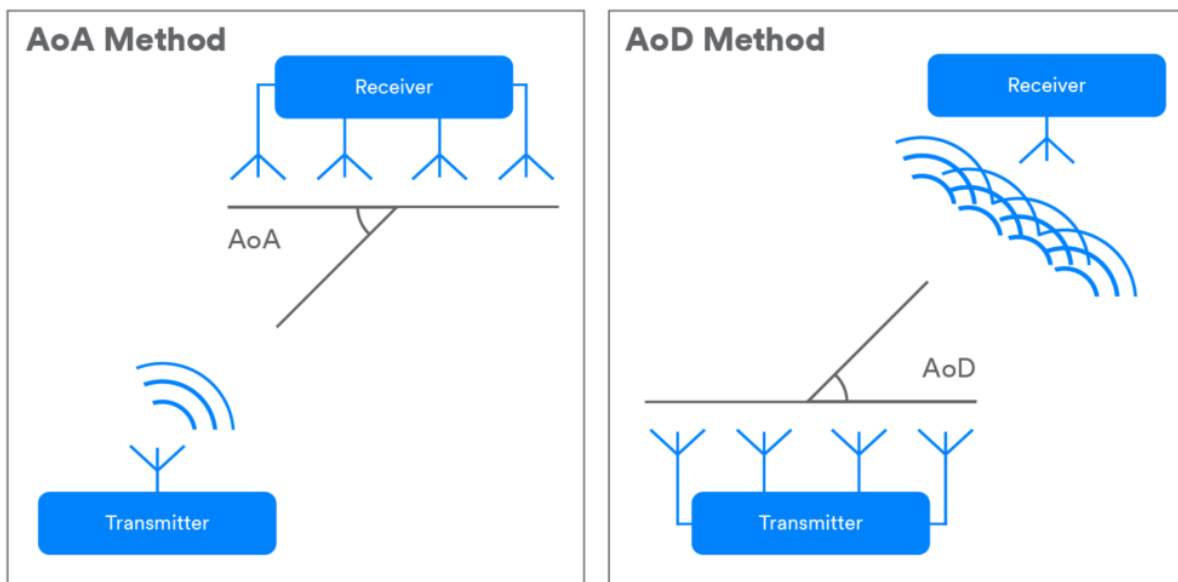


Figure 2 - Direction Finding using AoA and AoD

Phase measurements in the form of In-Phase and Quadrature (IQ) samples are passed from the Bluetooth controller to the application. IQ samples consist of pairs of phase and amplitude values which the application is able to use to calculate the direction in which the transmitter can be found.

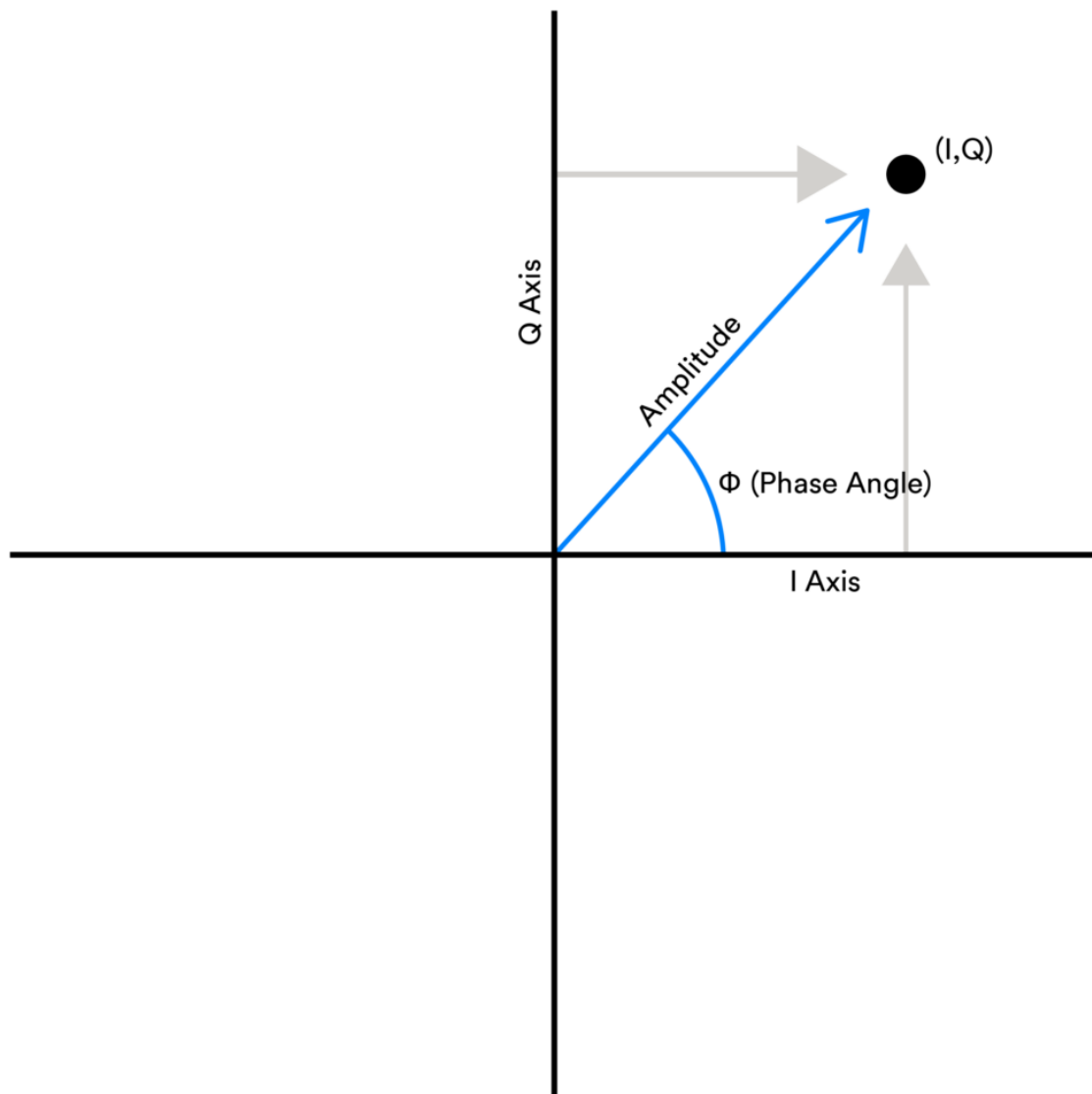


Figure 3 - IQ Sample

2.1.4 Bluetooth® Channel Sounding

The new Bluetooth® Channel Sounding feature makes it possible to create products that have the ability to calculate the distance between two Bluetooth devices with significantly better accuracy than could ever be produced using the RSSI and path loss first generation method. It works in an entirely different way and includes a variety of security safeguards that mitigate various types of risk.

It is expected that Bluetooth® Channel Sounding will benefit Find My solutions, digital key products, and many more Bluetooth connected devices.

2.2 An Introduction to Bluetooth® Channel Sounding

Before discussing Bluetooth® Channel Sounding in Bluetooth LE, this section will first present some of the basic theory behind the feature. Readers already familiar with the topic should skip to section 3, Bluetooth® Channel Sounding.

2.2.1 The Fundamental Properties of Radio Waves

Radio is a form of electromagnetic radiation and physicists often describe it in terms of waves. Radio waves have various basic properties, an understanding of which is important.

2.2.1.1 Amplitude and Wave Cycles

The amplitude of a radio wave corresponds to the energy that it carries, or, in more common terms, the signal strength. It oscillates above and below a central reference value. This above and below oscillation repeats regularly and periodically. A single transition up to the peak amplitude, down to the trough, and back up to the starting reference value is called a wave cycle. Figure 4 depicts two complete wave cycles, with amplitude on the vertical scale. The extent of the first wave cycle is highlighted.

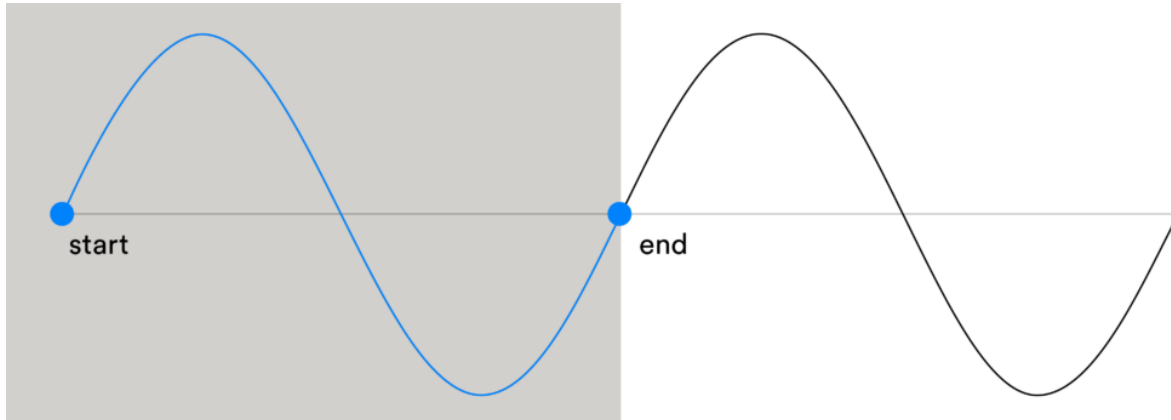


Figure 4 - Wave Cycle with amplitude on the vertical scale

2.2.1.2 Wavelength

A single wave cycle has a physical length. The wavelength is related to the frequency, and, in the case of Bluetooth technology, falls somewhere between about 12.0 cm and about 12.5 cm.

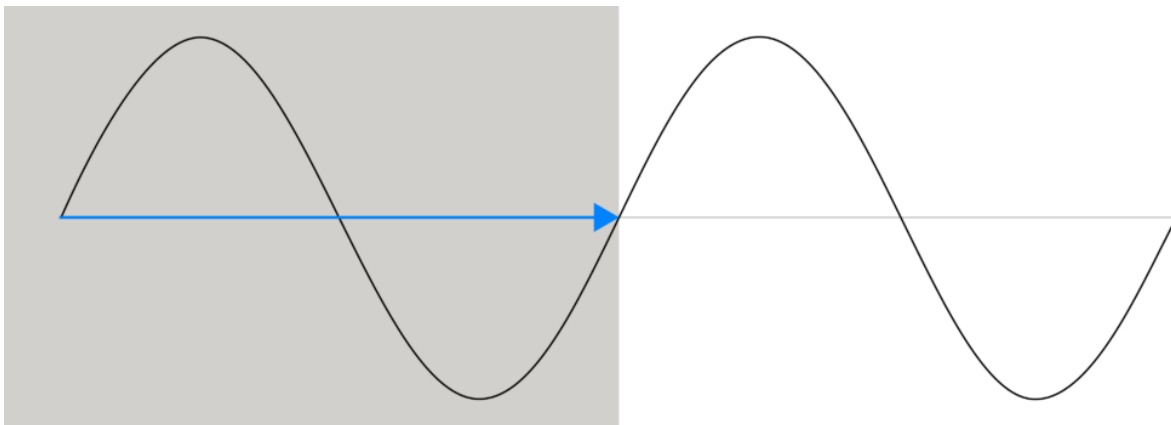


Figure 5 - Wavelength

2.2.1.3 Frequency

Radio travels at the speed of light in a vacuum². The number of complete wave cycles that pass over a fixed point in space in one second is called the frequency. Frequency is measured in hertz (Hz) where 1 Hz represents one wave cycle per second. A Bluetooth signal works at considerably higher frequencies measured in gigahertz (GHz).

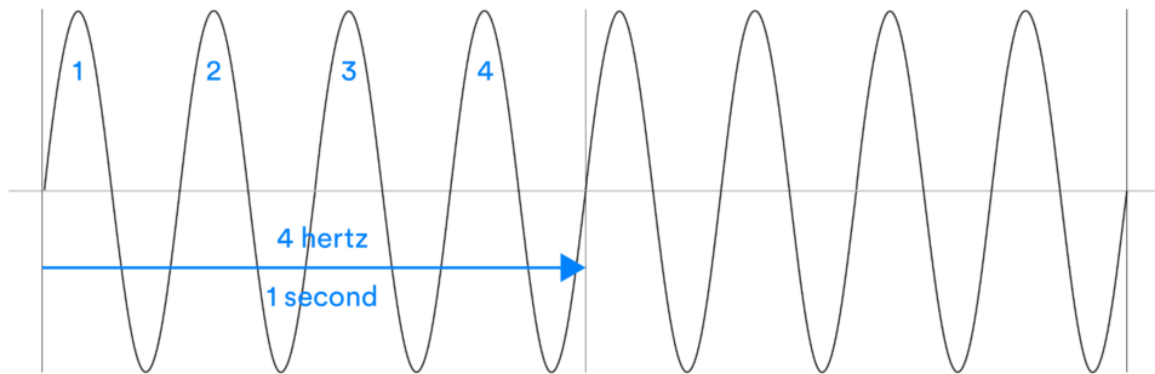


Figure 6 - Frequency

2.2.1.4 Phase

Points that sit somewhere within a single wave cycle are expressed by an angular measurement known as the phase. Phase values have a range of 0 - 360 degrees or 0 - 2π radians. Figure 7 illustrates the concept of phase with a number of phase values (expressed in radians) marked at appropriate points on the wave cycle.

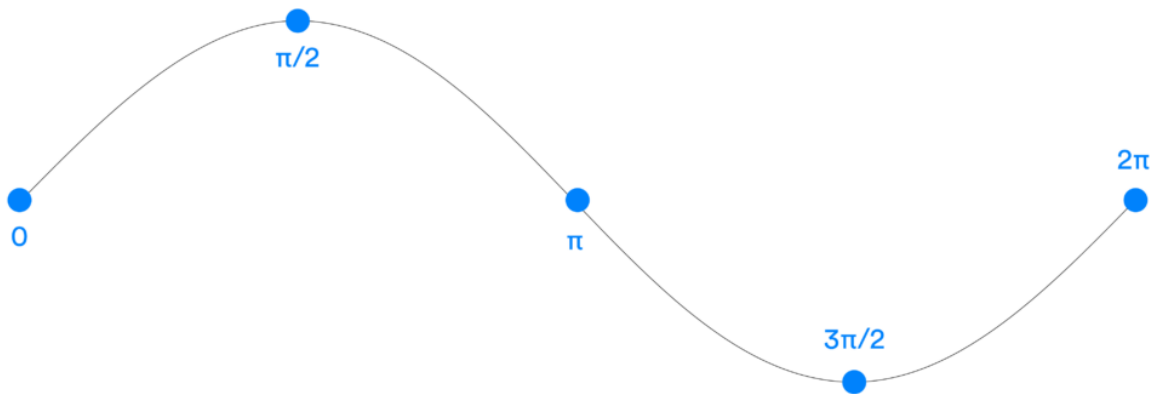


Figure 7 - Phase

2.2.1.5 The Mathematical Relationship Between Frequency and Wavelength

Frequency (f) and wavelength (λ) are inversely related to each other. The shorter the wavelength, the higher the frequency and vice versa. Furthermore, the relationship between these two variables and the speed of light (c) are defined by a set of simple formulae, allowing any one of the three quantities to be calculated from known values for the other two. The speed of light is a constant with a value of 299792458 m/s.

$$\lambda = \frac{c}{f}$$

$$f = \frac{c}{\lambda}$$

$$c = f * \lambda$$

Table 1 - Frequency and Wavelength Formulae

2.2.2 Distance Measurement Methods

The two most commonly used methods in wireless distance measurement technologies are Phase-Based Ranging (PBR) and Round-Trip Timing (RTT). The theory behind both methods will be outlined in this section.

2.2.2.1 Phase-Based Ranging (PBR)

2.2.2.1.1 Theory

It's easy to visualize distance as a function of the wavelength of a signal in terms of the number of wave cycles that are needed for the signal to stretch from a transmitter to a receiver.

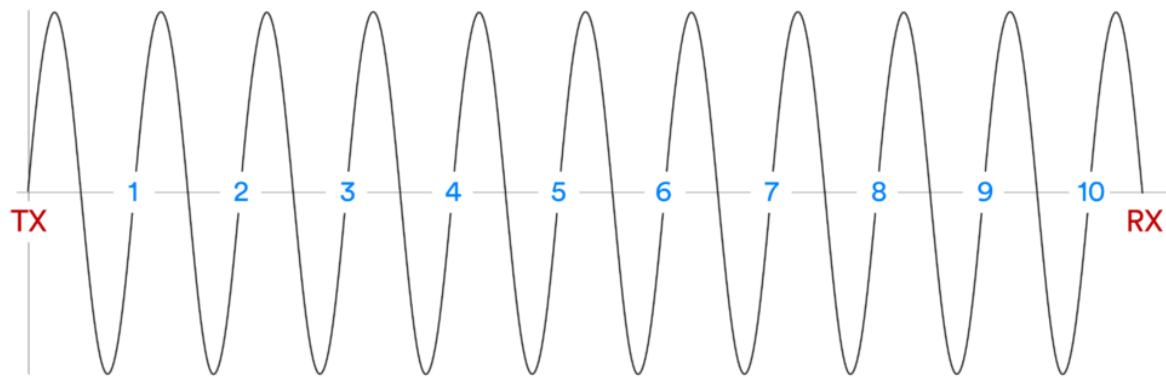


Figure 8 - Distance from wavelength and wave cycles

In Figure 8, the signal transmitted on the left of the illustration is clearly ten and a half wavelengths away from the receiver. If we know the frequency of the signal then we know the wavelength. And if we know the wavelength, then knowing the number of wave cycles, we can use multiplication to find the distance between the two devices.

If the transmission frequency is 2402 MHz for example, then the wavelength is 12.48095162 cm. This figure was arrived at by dividing the speed of light by the frequency.

The transmitting device has no way of knowing the number of wave cycles between its antenna and that of the receiver, however. So, the PBR method involves a technique which allows the deduction of the distance between transmitter and receiver based on other data. Here's how it works.

We'll refer to the device that wishes to calculate a distance measurement as Device A. The other device will be Device B.

- 1 Device A transmits a signal at a known frequency, f_1 . The initial phase of this signal is known to Device A, and, for the purpose of illustration, let's assume that this signal is transmitted with a phase of zero radians.
- 2 Device B receives the f_1 signal at its antenna and notes its phase, which we will refer to as the receive phase.
- 3 Device B then echoes the received signal back to Device A by transmitting on the same frequency, f_1 and, critically, ensuring that the initial phase of this transmission is exactly the same as the

receive phase of the signal received from Device A. This results in the return signal being a continuation of the signal from Device A in terms of phase and frequency.

- 4 Device A measures the receive phase of the signal arriving from Device B. We'll call this value Pf1.

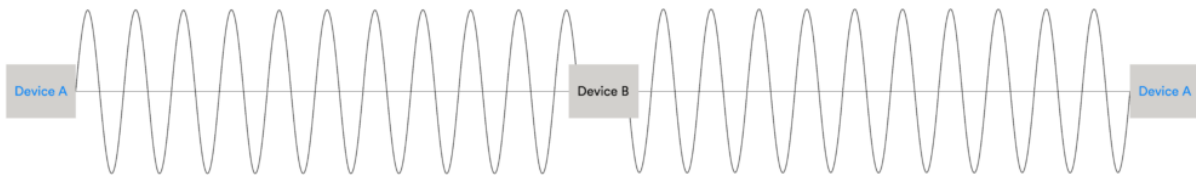


Figure 9 - Two-way ranging with frequency f1

Device A now chooses a new frequency, f2, and the four steps are repeated. The result of this second execution of the four steps is a new phase measurement made by Device A of the signal received back from Device B which we'll call Pf2.

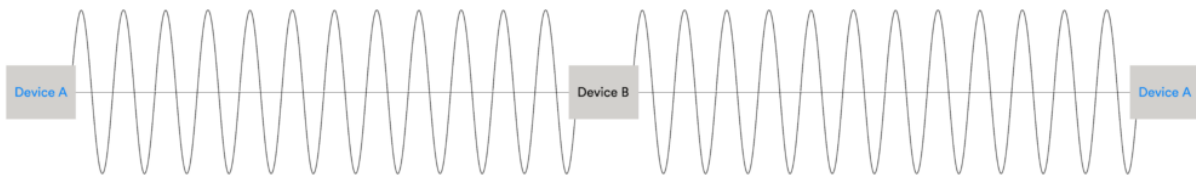


Figure 10 - Two-way ranging with frequency f2

Device A now calculates the difference between the phase values measured for each of f1 and f2, i.e., it calculates Pf2 - Pf1. Armed with the phase difference and the difference between the frequencies f1 and f2, it is now possible to calculate the distance using the following formula:

$$r = \left(\frac{c * (P_f2 - P_f1)}{2\pi (f_2 - f_1)} \right) \bmod \left(\frac{c}{(f_2 - f_1)} \right)$$

Where c is the speed of light, (Pf2 - Pf1) is the phase difference and (f2 - f1) is the frequency separation, and r is the two-way ranging distance between the two devices.

This approach, where the second device transmits a signal back to the originating device so that it can take phase measurements, is called two-way ranging.

The real-world can present challenges not reflected in this explanation of some basic theory. We'll encounter some of those challenges later in this section.

2.2.2.1.2 Worked Example

Let's try a simple worked example to see this in action. We'll use a rather artificial case where we already know the distance between the two devices so that we can see how the formula correctly arrives at the same result.

Figure 11 shows two devices, Device A and Device B, which are exactly 1.248095162 meters apart. Device A has transmitted a signal with a frequency of 2.402 MHz and a wavelength of 12.48095162 cm. By a truly remarkable coincidence, the two devices are therefore at a distance from each other of exactly ten times this wavelength.

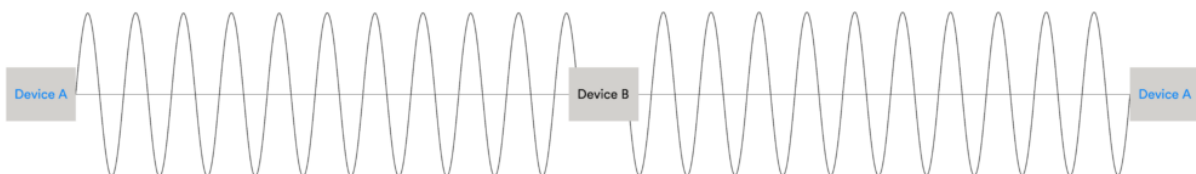


Figure 11 - Devices exactly ten f1 wave cycles apart

Given Device A transmits this signal with an initial phase of zero, and, since Device B is an exact multiple of the wavelength away, the receive phase at Device B is also zero. As shown, Device B transmits a signal back to device A, setting the initial phase to the same value of the receive phase as the signal it originally received so that we effectively have a continuation.

Figure 12 shows the second signal transmitted by Device A at frequency f_2 . This time the chosen frequency is a higher frequency than f_1 with a value of $f_2 = 2.432$ MHz. The initial phase at Device A is once again zero.

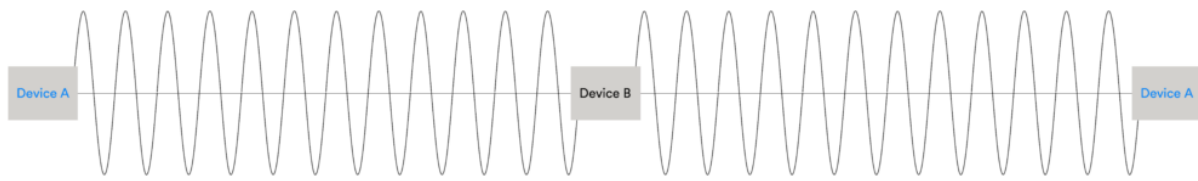


Figure 12 - Devices a little over ten f_2 wave cycles apart

The wavelength of f_2 is shorter than the wavelength of f_1 because f_2 has a higher frequency. This results in the receive phase at Device B being non-zero. In fact, it's 0.784744210 radians. By the time the signal has been retransmitted by Device B, with the same initial phase and received by Device A, its phase will be 1.56948842 radians.

Device A now has all that it needs to calculate the distance to Device B. The frequency difference is 30 MHz, and the phase difference is 1.56948842. Substituting these values in the formula for r , the calculated distance is 2.49 meters to two decimal places. But this is for the round trip from Device A to Device B and back again, so the actual distance between the two devices is half of that figure, i.e., 1.24 meters. This is the expected result and demonstrates how the formula for r , based on the speed of light and known phase and frequency separations of the two transmitted signals, can be used to accurately calculate the distance between the two devices.

There's a complication however and this is hinted at in the formula for phase and the modular division of $(2 * \pi)$. Phase values change as the distance increases, but they are periodic meaning that when a phase value reaches $(2 * \pi)$ radians it resets to zero and the same values start to be repeated. This can give rise to ambiguity in determining the distance between two devices since more than one distance could be implied by the same phase difference value. This is known as distance ambiguity.

Exactly when distance ambiguity is encountered depends on the frequency separation. In general, distance ambiguity occurs earlier with larger frequency differences. Luckily, the issue can be addressed by using PBR in conjunction with the second distance measurement method, Round-Trip Timing.

2.2.2.2 Round-Trip Timing (RTT)

2.2.2.2.1 Theory

The theory behind using round-trip timings to calculate the distance between two devices is very simple. Radio (RF) transmissions travel at the speed of light, a known constant. So if we can calculate the time it takes for a transmission to travel between two devices, we can calculate the distance. All we have to do is multiply the round-trip time by the speed of light.

For example, if an RF signal takes 20 nanoseconds to travel from Device A to Device B and back to Device A, simply multiplying the speed of light by 20 nanoseconds will give us a total bidirectional distance of just under six meters and thus the distance between the two devices is just under three meters.

$$\text{Bidirectional Distance } 2r = c * 0.00000002$$

where c is the speed of light (299792458 m/s) and 0.00000002 is the bidirectional time of flight (ToF) in seconds. This gives us the following result:

$$\begin{aligned} 2r &= 299792458 * 0.00000002 \\ &= 5.99584916 \end{aligned}$$

and therefore the distance between Device A and Device B is

$$2.99792458 \text{ meters}$$

But whilst this basic formula is correct, using it in the context of Bluetooth devices is a bit more complicated, and the theory as presented so far is incomplete.

The act of formulating and transmitting an RF signal takes time as does the act of receiving, processing, and transmitting the round-trip response. It might take a device something in the order of magnitude of 200 microseconds to formulate and transmit a packet, and, bearing in mind that radio waves can travel just under 300 meters in a single microsecond, these seemingly short time periods can be highly significant in the context of distance measurements.

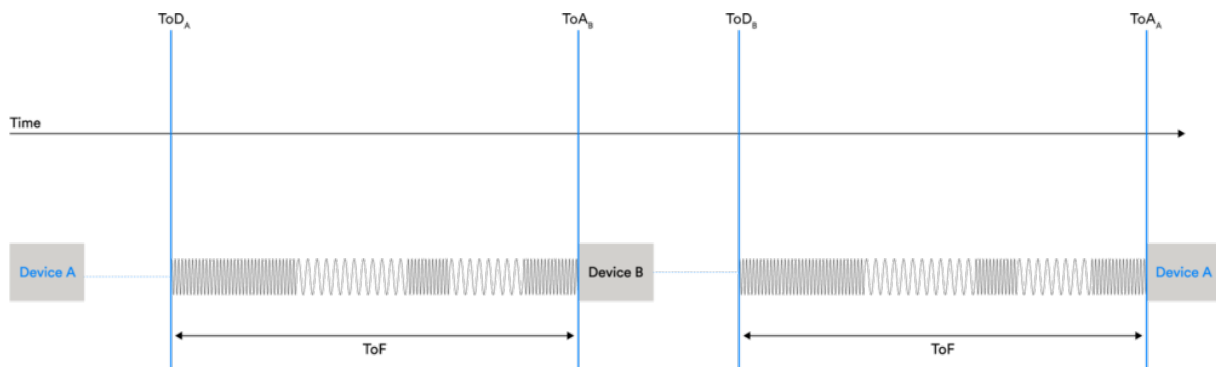


Figure 13 - RTT breakdown (Not to scale - signal content not representative)

Instant in Time	Explanation
ToD_A	Time of Departure from Device A. This is the time at which the signal is transmitted over the air by Device A.
ToA_B	Time of Arrival at Device B. This is the time at which the signal arrives at the antenna of Device B.
ToD_B	Time of Departure from Device B. This is the time at which Device B transmits over the air.
ToA_A	Time of Arrival at Device A. This is the time at which the signal from Device B is received at Device A's antenna.

The green dotted lines represent elapsed time during which neither of the two signals are in the air.

The round-trip time (RTT) can be expressed in terms of the timing instants depicted in Figure 13 as follows:

$$RTT = 2 * ToF = (ToA_A - ToD_A) - (ToD_B - ToA_B)$$

For Device A to calculate RTT, it needs to know the turnaround time at Device B (i.e., $ToD_B - ToA_B$). In theory, there are a number of ways in which this could work. In practice, the simplest solution is for a fixed turnaround period to be agreed by Device A and Device B in advance. Device B must then guarantee to complete its processing and, exactly as that turnaround period expires, transmit its response. Device A then uses that pre-agreed value for $(ToD_B - ToA_B)$.

2.2.2.3 Real-World Challenges

The theory presented for both the PBR and RTT methods of distance measurement is sufficient for gaining an initial insight into the topic and, in a purely theoretical context, it is complete. However, in the real world, accurate distance measurement is more complicated. There are several challenges which must be addressed if satisfactory results are to be produced by real devices used in real-world situations.

Examples of the types of challenges that a wireless distance measurement technology should address are:

- The complications arising from multi-path propagation of radio signals
- The accuracy and stability of the frequency of generated signals
- The stability of internal clocks and the accuracy and resolution of timestamps
- Distance ambiguity in phase-based ranging
- Security

In the remainder of this paper, we'll learn about high-accuracy distance measurement in Bluetooth technology and gain an appreciation of how the technology has been designed to work effectively in the face of real-world issues such as these.

3. Bluetooth® Channel Sounding

3.1 Overview

Bluetooth® Channel Sounding offers the potential for products to achieve much higher accuracy distance measurements than has previously been possible. How accurate measurements are depends on environmental conditions and how the Bluetooth® Channel Sounding feature is harnessed by the application layer. It also depends on implementation choices, details of which fall outside of the Bluetooth Core Specification but which can improve the quality of the raw data used in calculations.

Bluetooth® Channel Sounding provides applications with a flexible toolkit for distance measurement with a number of quite different configurations possible. Both the Phase-based Ranging (PBR) and Round-Trip Timing (RTT) distance measurement methods are supported in the specification. In most cases, it is expected that PBR will be used as the primary and most accurate method of distance measurement with RTT used alongside it to provide extra security.

PBR, as used by Bluetooth® Channel Sounding, can measure distances up to about 150 meters before distance ambiguity arises. By using RTT in addition to PBR, applications can identify and eliminate distance ambiguity and therefore measure longer distances.

Applications may place different levels of priority over issues like accuracy, security, latency, and power consumption. The configurability of the Bluetooth® Channel Sounding feature provides applications with control or influence over many of the system's key capabilities and behaviors so that its operation is focused on the right priorities for the application using it.

In this section we will proceed to examine the Bluetooth® Channel Sounding feature and the core Bluetooth stack capabilities upon which it depends.

3.2 Architecture

3.2.1 Device Roles

The Bluetooth® Channel Sounding feature defines two device roles. The first is the Initiator, and the second is the Reflector.

The Initiator is the device that wishes to calculate the distance from itself to another device. The other device is the Reflector.

Either the Initiator or the Reflector can kick off the Bluetooth® Channel Sounding procedure, details of which will be covered later in this paper.

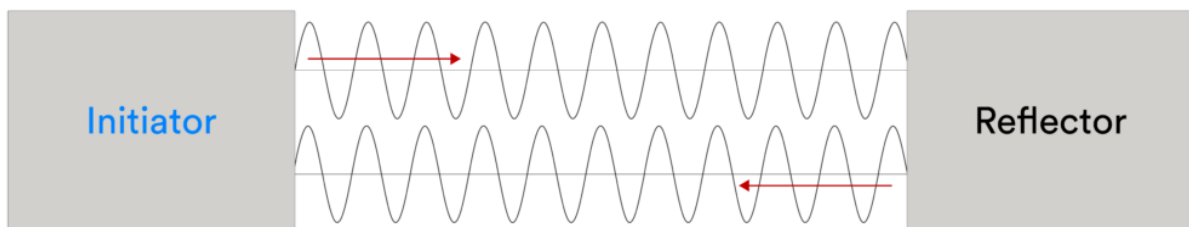


Figure 14 - Roles

3.2.2 Topology

Bluetooth® Channel Sounding takes place in a one-to-one topology with communication taking place between one device in the Initiator role and one device in the Reflector role.

It should be noted that the Bluetooth® Channel Sounding Initiator role may be assumed by either the device acting in the Link Layer LE Central role or as the LE Peripheral. The same applies to the

Bluetooth® Channel Sounding Reflector role which may be assumed by either the LE Central device or the LE Peripheral device.

3.2.3 Antenna Arrays

Devices that use Bluetooth® Channel Sounding may include an antenna array. This provides a series of alternate paths for the exchange of the Bluetooth® Channel Sounding transmissions that are used for phase-based ranging and can improve the accuracy of distance measurements by lessening the impact of multi-path propagation.

3.2.4 Applications

Bluetooth® Channel Sounding requires that distances are calculated by the application layer using data provided by the Bluetooth controller. That data is acquired by the controller during the execution of the Bluetooth® Channel Sounding procedure and is a result of signal exchanges and low-level measurements made at each device. Data is passed to the application layer in HCI events.

The application layer is also responsible for providing the Bluetooth controller with configuration choices and preferences which are used in establishing a Bluetooth® Channel Sounding configuration that is supported by and suitable for the applications on the two devices.

For two devices to be able to participate in a system with one device in the Initiator role and the other in the Reflector role, both must have a Bluetooth LE controller which supports the Bluetooth® Channel Sounding feature.

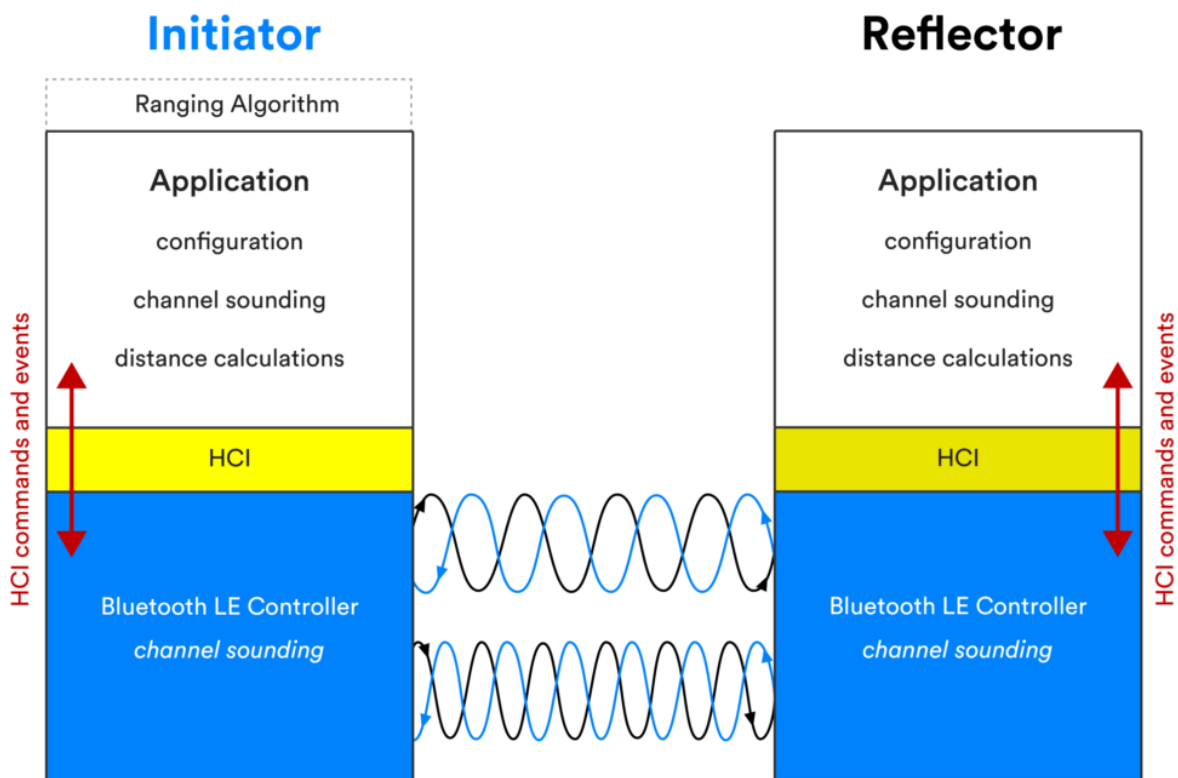


Figure 15 - Bluetooth® Channel Sounding applications and the Bluetooth stack

3.2.5 Data Transport Architecture

The Bluetooth Core Specification defines the architecture of Bluetooth technology from a number of perspectives. In the first perspective, a generalized data transport architecture is defined.

With reference to definitions in the Bluetooth Core Specification, the terms in Figure 16 are described as follows:

- **L2CAP** is the Logical Link Control and Adaptation Protocol. An L2CAP Channel is a logical connection at the L2CAP level between two devices that serves a single application or higher layer protocol.

- A **Logical Link** is “The lowest architectural level used to offer independent data transport services to clients of the Bluetooth system”.
- A **Logical Transport** deals with issues such as transmit and receive routines, flow control mechanisms, acknowledgment protocols, and link identification. A Logical Transport can be synchronous, asynchronous, or isochronous.
- A **Physical Link** is a connection between devices established at the level of the Link Layer. The Link Layer is one of the layers of the Bluetooth protocol stack.
- **Physical Channels** define patterns of occupancy of RF carriers by one or more communicating devices.
- **Physical Transports** define generally applicable issues, such as over-the-air packet structures and modulation schemes which are used to encode digital data for transmission using radio signals as the carrier.

The generic data transport architecture applies to both Bluetooth LE and Bluetooth Basic Rate/Enhanced Data Rate (BR/EDR).

Figure 17 shows a subset of the Data Transport Architecture. Highlighted in blue are a new Physical Channel type and a new Physical Link type that have been defined for Channel Sounding.

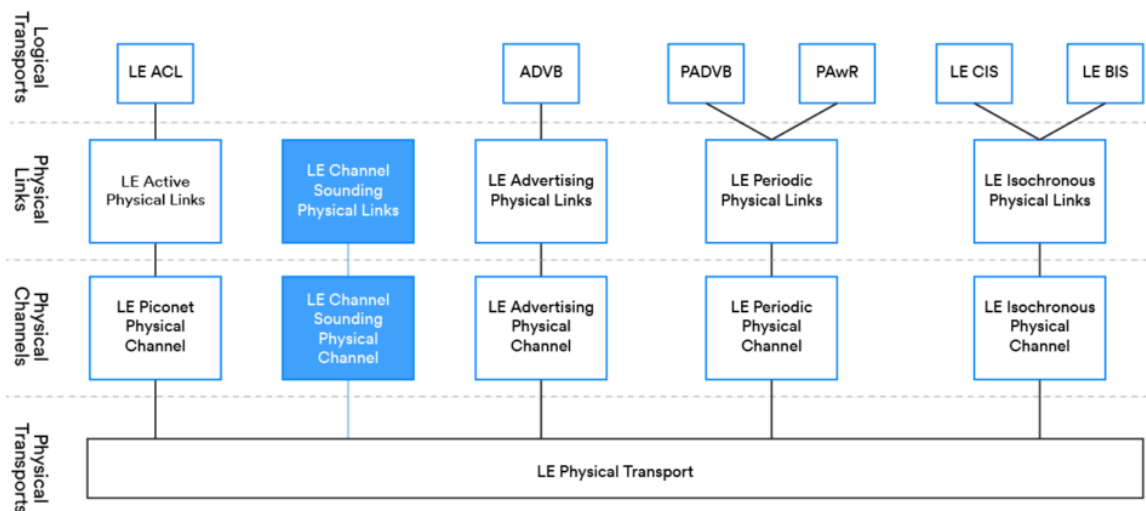


Figure 17 - CS and the Data Transport Architecture

No Logical Transport type or Logical Link type is associated with the LE Channel Sounding physical link.

3.2.6 Channel Sounding in the Bluetooth LE Stack

The more comprehensive way of defining Bluetooth LE is in terms of the full protocol stack and its layers. Much of the Bluetooth Core Specification is dedicated to defining each layer. Figure 18 depicts the Bluetooth LE stack.

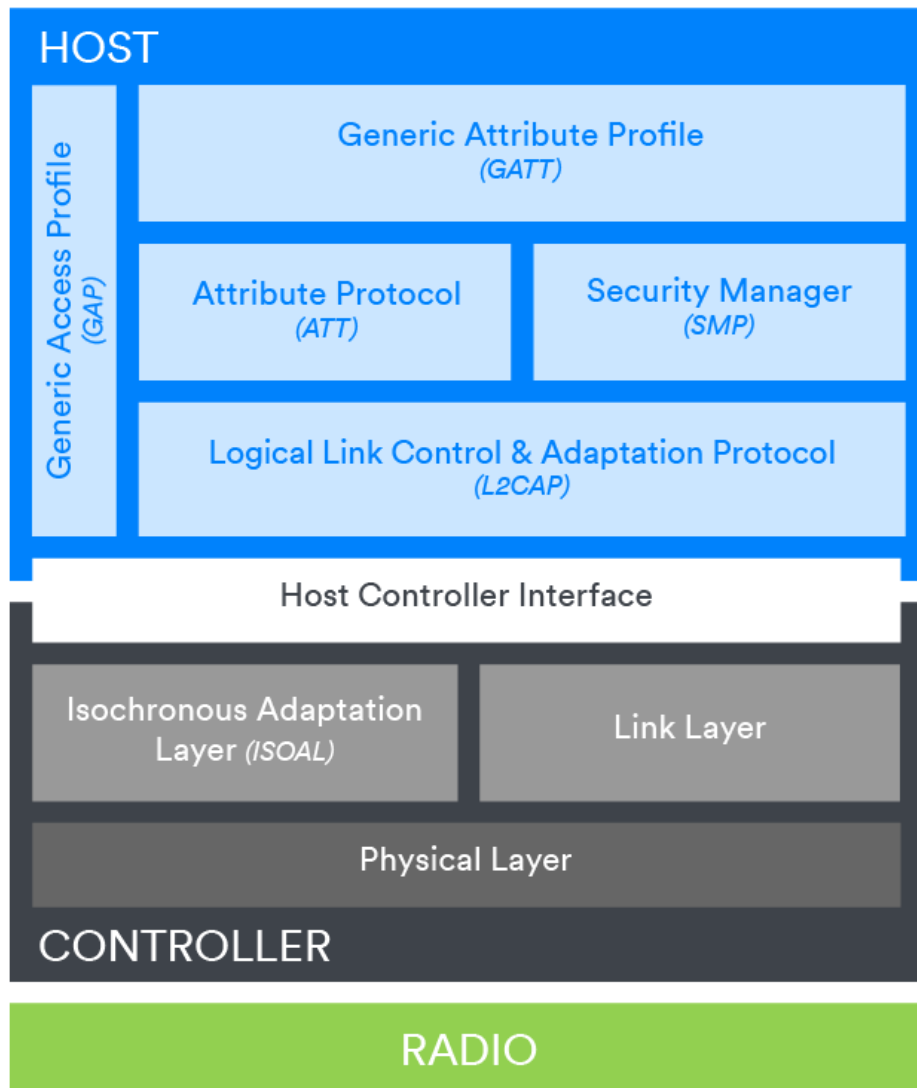


Figure 18 - The Bluetooth LE stack

A summary of the responsibilities of each layer of the Bluetooth LE stack is contained within Table 2.

Layer	Key Responsibilities
Generic Access Profile (GAP)	Defines operational modes and procedures that may be used in a non-connected state, such as how to use advertising for connectionless communication and device discovery. Defines security levels and some user interface standards.
Generic Attribute Profile (GATT)	Defines high-level data types known as services, characteristics, and descriptors in terms of underlying attributes in the attribute table.
Attribute Protocol (ATT)	A protocol used for the discovery and use of data held by the server in a logical data structure known as the attribute table.
Security Manager Protocol (SMP)	A protocol used during the execution of security procedures such as pairing.
Logical Link Control and Adaptation Protocol (L2CAP)	Provides data channel multiplexing services over RF connections, segmentation and reassembly of large SDUs, and enhanced error detection and retransmission facilities.
Host Controller Interface (HCI)	Provides an interface for bi-directional communication of commands and data between the host component and the controller.
Isochronous Adaptation Layer (ISOAL)	Allows different frame durations to be used by devices using isochronous channels.
Link Layer	Defines air interface packet formats, bit stream processing procedures such as error checking, a state machine and protocols for over-the-air communication, and link control. Defines several distinct ways of using the underlying radio for connectionless, connection-oriented, and isochronous communication known as logical transports.

Layer	Key Responsibilities
Physical Layer	Defines all aspects of Bluetooth technology that are related to the use of radio (RF), including modulation schemes, frequency bands, channel use, transmitter, and receiver characteristics. Three combinations of Physical Layer parameters are defined and are referred to as the LE 1M, LE 2M, and LE 2M 2BT PHYs. LE 2M 2BT was defined for the first time in the version 6.0 of the Bluetooth Core Specification and may only be used with Bluetooth® Channel Sounding. A further PHY, LE Coded, is defined. Despite the name, LE Coded uses the same Physical Layer parameters as LE 1M but applies forward error-correction coding and pattern mapping in the Link Layer.

Table 2 - Summary of the key responsibilities and features of each layer of the Bluetooth LE stack

The Physical Layer, Link Layer, Host Controller Interface, and Generic Access Profile sections of the Bluetooth Core Specification have all been impacted by the introduction of Bluetooth® Channel Sounding. Section 4 explains this further.

3.3 Bluetooth® Channel Sounding Control Procedures

Before Bluetooth® Channel Sounding can be started, the device in the link layer LE Central role must connect to the device in the link layer LE Peripheral role. Security is then started on the LE-ACL connection that is established so that it can provide a secure transport for the exchange of various link layer Protocol Data Units (PDUs) during several procedures that are concerned with preparing for and then initiating Bluetooth® Channel Sounding.

The main procedures which prepare for and then initiate Bluetooth® Channel Sounding are:

- 1 Security Start
- 2 Capabilities Exchange
- 3 Configuration
- 4 Start

Not all of these procedures are mandatory, depending on issues such as whether or not the two devices have previously exchanged information which may have been cached. A possible sequence of procedures and associated PDUs is shown in Figure 19.

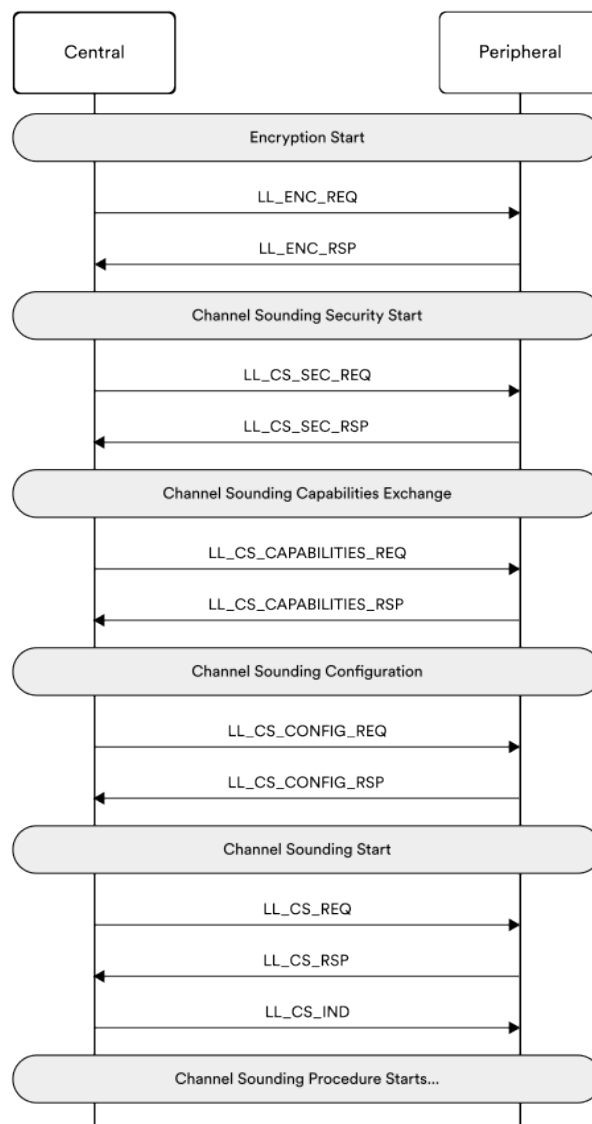


Figure 19 - A possible CS initiation procedure sequence

3.3.1 Bluetooth® Channel Sounding Security Start

Bluetooth® Channel Sounding has its own security capabilities that are distinct from those associated with the LE-ACL connection over which the initialization procedures are executed. The Bluetooth® Channel Sounding Start Procedure allows the two devices to securely exchange parameters that are later used in Bluetooth® Channel Sounding security functions.

The Bluetooth® Channel Sounding Security Start procedure starts with the LE Central device generating three random numbers and sending them to the LE Peripheral device in a `LL_CS_SEC_REQ` PDU. The LE Peripheral device generates three random numbers of its own, governed by the same rules as the Central's random numbers and sends them back to the Central in a `LL_CS_SEC_RSP` PDU.

The random numbers generated by each device are named and described in Table 3.

Name	Description	Length (bits)
CS_IV_C	Initialization Vector generated by the Central.	64
CS_IN_C	Instantiation Nonce generated by the Central.	32
CS_PV_C	Personalization Vector generated by the Central.	64
CS_IV_P	Initialization Vector generated by the Peripheral.	64
CS_IN_P	Instantiation Nonce generated by the Peripheral.	32
CS_PV_P	Personalization Vector generated by the Peripheral.	64

Table 3 - CS Security Parameters

When both devices are in possession of both sets of Bluetooth® Channel Sounding security parameters, the values of each Central/Peripheral pair are concatenated by the respective link layers. This results in both devices possessing the same values for the three Bluetooth® Channel Sounding security parameters CS_IV, CS_IN and CS_PV.

3.3.2 Bluetooth® Channel Sounding Capabilities Exchange

The Bluetooth® Channel Sounding capabilities of two devices may vary significantly and in order that a mutually supported configuration can be arrived at before starting, the two devices must each be in possession of information about the capabilities of the other device.

The exchange of capabilities is achieved by one device sending its details in a LL_CS_CAPABILITIES_REQ PDU and the other responding with its details in a LL_CS_CAPABILITIES_RSP PDU. A device may cache capabilities data previously received by a device and therefore elect to not exchange capabilities with the other device. Either device may initiate this procedure, however.

Examples of ways in which capabilities may vary include PHY support, RTT accuracy, Bluetooth® Channel Sounding modes³ supported, attack detection support, and the maximum number of antenna paths supported.

3.3.3 Bluetooth® Channel Sounding Configuration

This procedure involves the exchange of LL_CS_CONFIG_REQ and LL_CS_CONFIG_RSP PDUs. In essence, using the capabilities previously exchanged, this procedure then allows devices to select the specific configuration that will be used.

Multiple configuration parameter sets may be maintained. Each such configuration is assigned an identifier by the Host. This identifier must be unique amongst the identifiers used by this pair of devices and may be used to reference a given parameter set during link layer procedures.

The application on the device which transmits the LL_CS_CONFIG_REQ PDU is able to select which of the Initiator or Reflector roles it wants to assume. The other device responds with LL_CS_CONFIG_RSP and must assume the other role.

3.3.4 Mode-0 FAE Table Request

Fractional Frequency Offset Actuation Error (FAE) is a measure of the difference between a generated frequency and the expected or requested frequency, expressed in Parts Per Million (ppm). All devices have some degree of inaccuracy in this respect, and, typically, its magnitude will vary according to the RF channel in use.

For the purposes of achieving distance measurement results that are as accurate as possible, a device which supports Bluetooth® Channel Sounding may have a table of data called the Mode-0 FAE Table. The table contains FAE values for each channel and is set up during the manufacturing process.

The Mode-0 FAE Table Request procedure allows an Initiator to request a Reflector's mode-0 FAE table. This involves the Initiator transmitting a LL_CS_FAE_REQ PDU to which the Reflector replies with a LL_CS_FAE_RSP PDU containing its FAE table.

Once acquired, the FAE table may be stored for future use with the same Reflector so that this procedure need only be executed once for a given device pair.

3.3.5 Bluetooth® Channel Sounding Start

When Bluetooth® Channel Sounding security has been started, devices are in possession of information about each others' capabilities, the Initiator has the Reflector's mode-0 FAE table (if it has one) and the devices have agreed on a suitable configuration, then the Channel Sounding Start procedure may be initiated. This is accomplished via LL_CS_REQ, LL_CS_RSP and LL_CS_IND PDUs.

The LL_CS_REQ and LL_CS_RSP PDUs include proposed timing and structural parameters from each device. These parameters govern the way that time is divided and how it is made use of during

Bluetooth® Channel Sounding. A LL_CS_IND PDU is sent by the device in the Central role after receiving a LL_CS_REQ or LL_CS_RSP PDU from the Peripheral. The LL_CS_IND indicates that Bluetooth® Channel Sounding should now start and contains parameter values that are acceptable to both devices based on the proposals contained within the previous exchanges of PDUs.

3.3.6 Bluetooth® Channel Sounding

The Bluetooth® Channel Sounding procedure starts after the Bluetooth® Channel Sounding Start procedure has completed. This is the mechanism by which the two devices exchange RF signals for the purpose of taking measurements that can be used by an application for distance calculations.

3.4 Events, Subevents, and Steps

3.4.1 LE-ACL Connections and Time Division

In an ACL connection, packets can be transmitted during a connection event. The timing of connection events is based on the value of the connection interval parameter for that ACL connection. During a connection event, the Central and the Peripheral devices each take turns to transmit a packet with the Central transmitting first and the Peripheral responding. Depending on other connection parameters, the Peripheral may be permitted to only respond to a subset of packets and the Central may be permitted to only transmit during a subset of events.

An LE-ACL connection is used during the initiation procedures of Bluetooth® Channel Sounding as described in 3.3 Bluetooth® Channel Sounding Control Procedures.

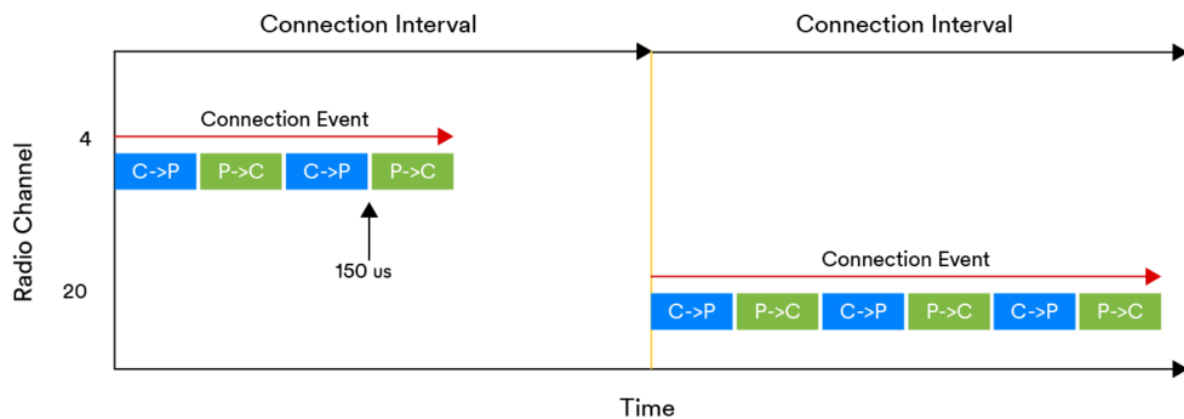


Figure 20 - Connection Events and Intervals in an LE-ACL Connection

3.4.2 Time Division

3.4.2.1 Structure

Bluetooth® Channel Sounding takes place in a series of procedures. Each procedure consists of a number of CS Events, and each CS Event is further partitioned into CS Subevents. The final subdivision of time within this hierarchical scheme is the CS Step. It is within steps that packets or tones are transmitted and received. Figure 21 depicts this structural scheme for the partitioning of time by way of an example.

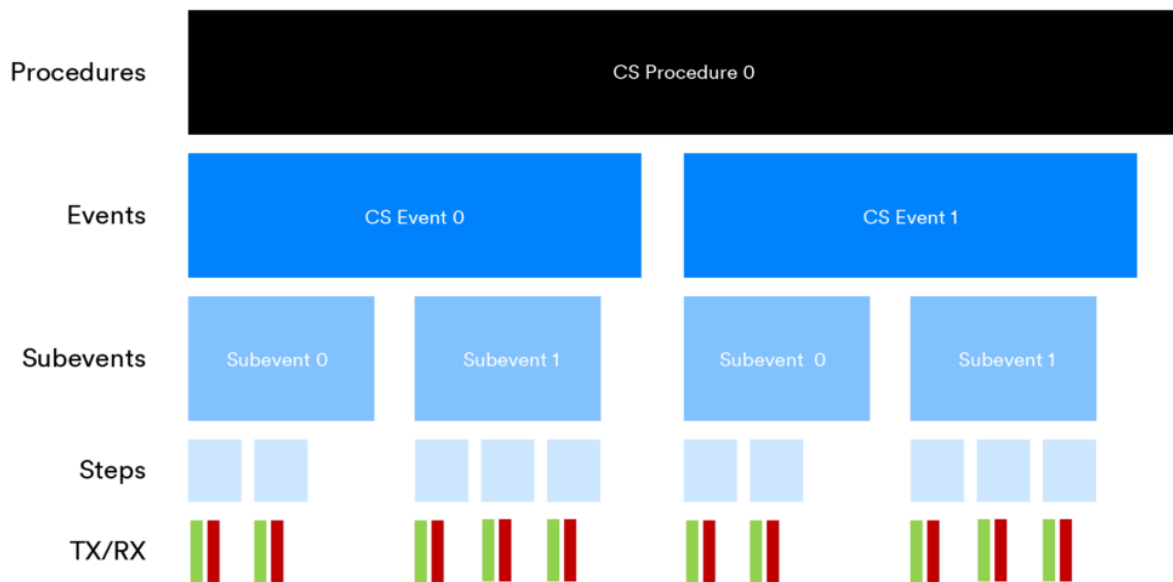


Figure 21 - Structure of Bluetooth® Channel Sounding procedures in an example configuration

There are a number of parameters that allow the structural aspects of Bluetooth® Channel Sounding procedures to be controlled. Some of the key configurable variables are shown in Table 4.

Configurable Variable	Range/Value	Description
Number of CS procedure repetitions	0 to 65535	The number of CS procedure repetitions to execute before Bluetooth® Channel Sounding is terminated. A value of 0 indicates that CS procedures should run until terminated via the Bluetooth® Channel Sounding Procedure Repeat Termination procedure.
Number of subevents per event	1 to 16	The number of subevents anchored off the same ACL event.
Subevent Interval	0 or in the range 625 us to 40959.375 ms.	Time interval between the beginning of a CS subevent and the beginning of the next CS subevent within the same CS event. 0 means no division into subevents.
Duration of each subevent	Variable	The duration of each subevent.
Number of steps per subevent	2 to 160	Randomly selected from a configured range. There are a maximum of 256 steps per procedure.

Table 4 - Example Bluetooth® Channel Sounding configuration parameters

3.4.2.2 Timing

The timing, duration, and scheduling of procedures, events, subevents, and steps is controlled by a number of parameters which are configured during the Bluetooth® Channel Sounding Configuration and Bluetooth® Channel Sounding Start procedures.

All procedure, event, subevent, and step start times are directly or indirectly anchored to a selected connection event in the underlying LE ACL connection. In the first Bluetooth® Channel Sounding procedure instance, its first event and subevent all start at the same time, scheduled to occur at an offset from the selected connection event anchor point. The first step occurs at an offset from the start of the first subevent called T_{FCS} . T_{FCS} has a value in the range 15 μ s to 150 μ s, and the period it covers is used to change the frequency by hopping.

Both procedures and events occur at intervals whose value is expressed in terms of a number of ACL connection intervals. Figure 22 shows an example where the procedure interval has a value of 4 and the event interval a value of 2.

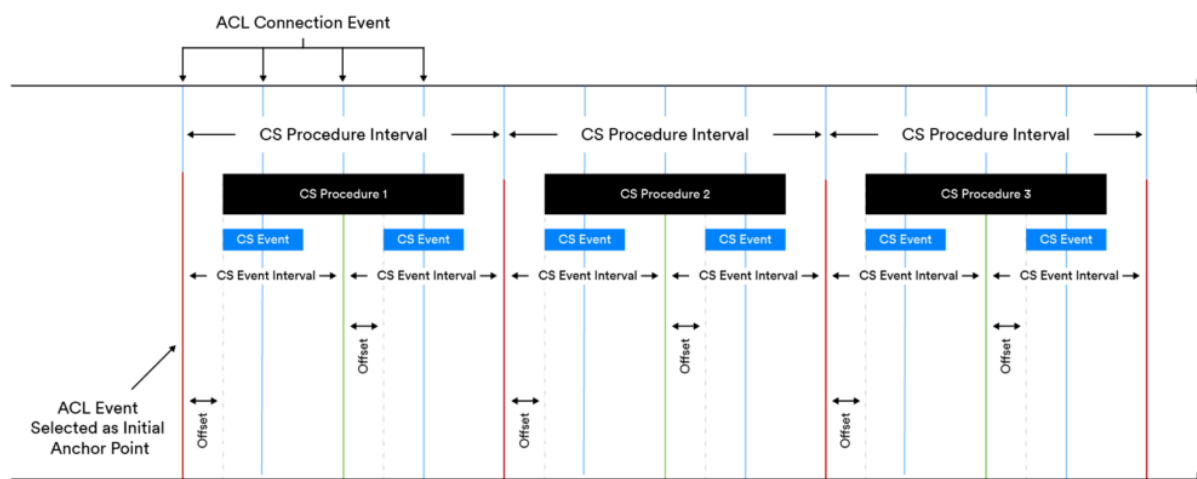


Figure 22 - Procedure and Event Scheduling with procedure interval = 4 and event interval = 2

The first subevent in each event starts at the same time as the event, offset from the relevant ACL connection event. The number of subevents per event is a configuration parameter and subevents occur once per subevent interval as shown in Figure 23.

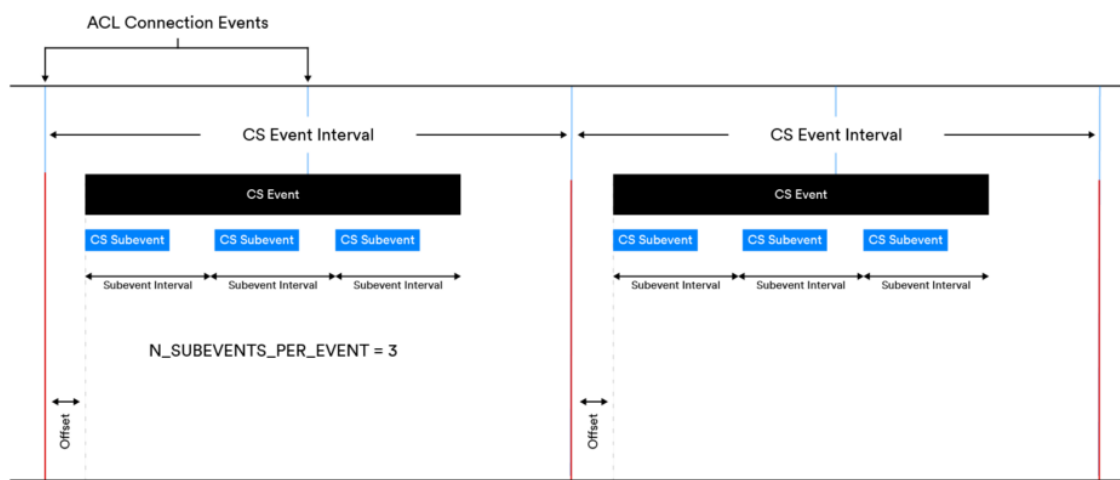


Figure 23 - Example of CS subevent within CS event scheduling

Each subevent includes at least two steps. This can vary from subevent to subevent, depending on how channel sounding is being used by the application. Steps can vary in duration, again, depending on configuration. The scheduling of steps and the RF transmission and reception slots assigned to them is subject to meticulous timing rules, further details of which can be found in the Bluetooth Core Specification.

3.5 Bluetooth® Channel Sounding Steps

3.5.1 About Steps

It is within steps that an exchange of RF signals between Initiator and Receiver takes place. Depending on the channel sounding method or methods that the application layer has opted to use (PBR and/or RTT), details vary.

In general, steps are either concerned with calibration or with the acquisition of low-level measurements that can be used by the application layer in a distance measurement algorithm.

3.5.2 Packets and Tones

When RTT is in use, packets of a type called CS_Sync are exchanged by Initiator and Reflector.

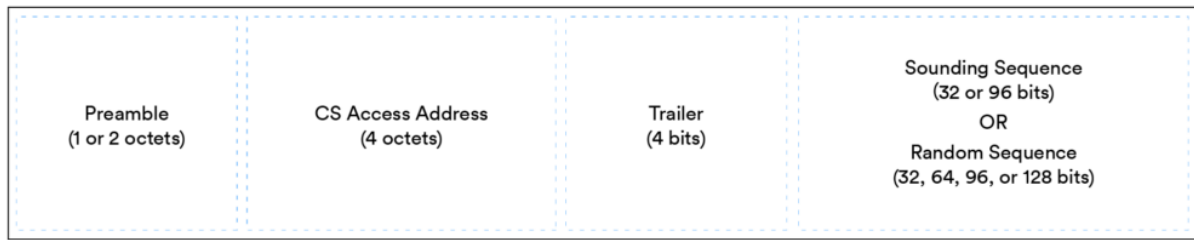


Figure 24 - The CS_Sync Packet

The inclusion of a Sounding Sequence or Random Sequence at the end of a CS_Sync packet is optional. CS_Sync packets can be transmitted using either the LE 1M, LE 2M or LE 2M 2BT PHY. The GFSK⁴ modulation scheme is used as with other Bluetooth LE packets.

When PBR is in use, signals known as CS Tones are exchanged by Initiator and Reflector. These signals use Amplitude Shift Keying (ASK) to create a symbol whose frequency is fixed for a specified period of time.

3.5.3 Step Modes

Steps have an associated mode which determines the goal of the step and the type of activity that takes place within it. Four modes are defined and are designated mode-0, mode-1, mode-2, and mode-3.

3.5.3.1 Mode-0

Mode-0 is concerned with calibration. All devices will exhibit some degree of clock drift and inaccuracy in frequency generation. This is an issue for both the RTT and PBR distance measurement methods.

The purpose of a mode-0 step is to allow the Initiator to measure the amount by which the frequency of signals transmitted by the Reflector differ from those generated by the Transmitter.

The Initiator transmits a CS_Sync packet on a selected channel and frequency. The Reflector replies with a CS_Sync packet and a CS Tone. Both are required to be transmitted on the same frequency as the signal received from the Initiator.

On receiving the response signal from the Reflector, the Initiator calculates a value called the Fractional Frequency Offset (FFO). The calculation of FFO involves the frequency of the tone received from the Reflector and the Reflector's mode-0 FAE table. FFO is later used in calculations to compensate for the differences between the two devices and improve the accuracy of the results.

Figure 25 shows the transmission of a CS_Sync packet by the Initiator followed by the CS_Sync and CS Tone sent in response by the Reflector. The duration of various time slots are indicated by symbolic names with the following meanings:

Symbol	Meaning
T_SY	Time for synchronization sequence. Duration depends on CS_Sync packet length and the PHY used.
T_RD	Time for transmission ramp down. This is 5 μ s and is used by the transmitter to remove energy from the RF channel.
T_IP1	Time for interlude period between the end of the Initiator's transmission and the start of the transmission by the Reflector. Durations vary between 10 μ s and 145 μ s and are determined in the capabilities exchange procedure.
T_GD	Guard time. Always 10 μ s in duration.
T_FM	Time for frequency measurement. Always 80 μ s in duration for step mode-0.

Table 5 - Time slot parameters

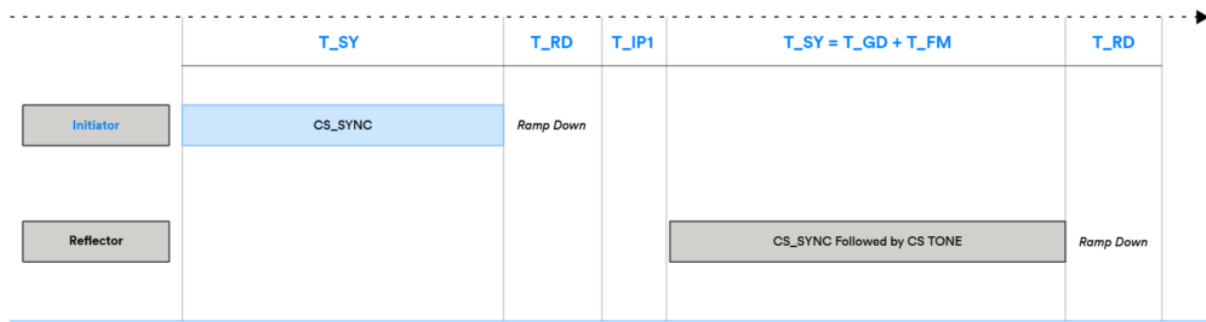


Figure 25 - Mode-0 transmissions and time slots

Support for mode-0 steps is mandatory.

3.5.3.2 Mode-1

In a mode-1 step, the round-trip timing (RTT) of a CS_Sync packet sent from an Initiator to a Reflector and back again is calculated.

A timestamp is recorded by the Initiator when transmitting the initial CS_Sync packet and is known as the Time of Departure (ToD). The Initiator records a second timestamp on receiving the CS_Sync packet sent back by the Reflector. This is known as the Time of Arrival (ToA).

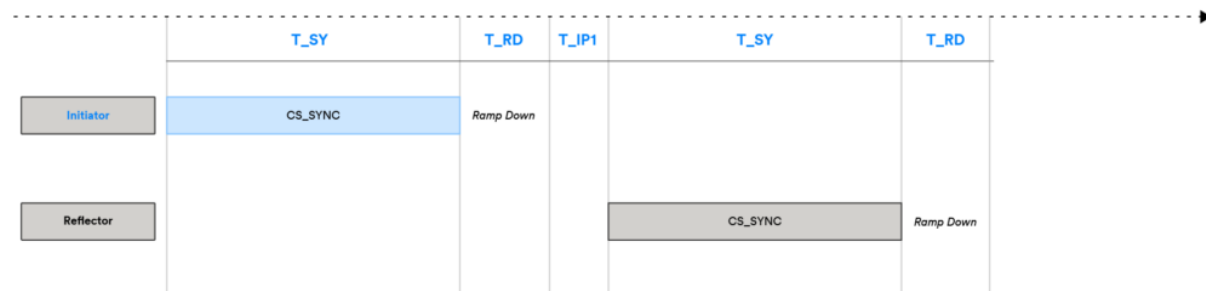


Figure 26 - Mode-1 transmissions and time slots

The interlude period, T_{IP1} is of a known fixed length that is sufficient in duration for the Reflector to prepare and then transmit its packet. The use of a pre-agreed, fixed period in this part of the exchange means that the Initiator knows the turnaround time at the Receiver and can use this in its RTT calculation.

Support for mode-1 steps is mandatory.

3.5.3.3 Mode-2

The purpose of a mode-2 step is to support phase-based ranging (PBR).

A mode-2 step starts with the Initiator transmitting a CS Tone on a selected channel and via each available antenna path. After a ramp down time and interlude period, the Reflector replies with a CS Tone, selecting the same frequency as the tone received from the Initiator and over each of its antenna paths. Figure 27 illustrates the exchange. Time slot duration involves the terms described in Table 5 and the additional terms defined in Table 6.

Symbol	Meaning
T_{SW}	Time period reserved for antenna switching.
T_{PM}	Time for the transmission of a phase measurement tone.
T_{IP2}	Time for interlude period between CS Tones.
N_{AP}	Number of antenna paths.

Table 6 - Additional step mode-2 timing parameters

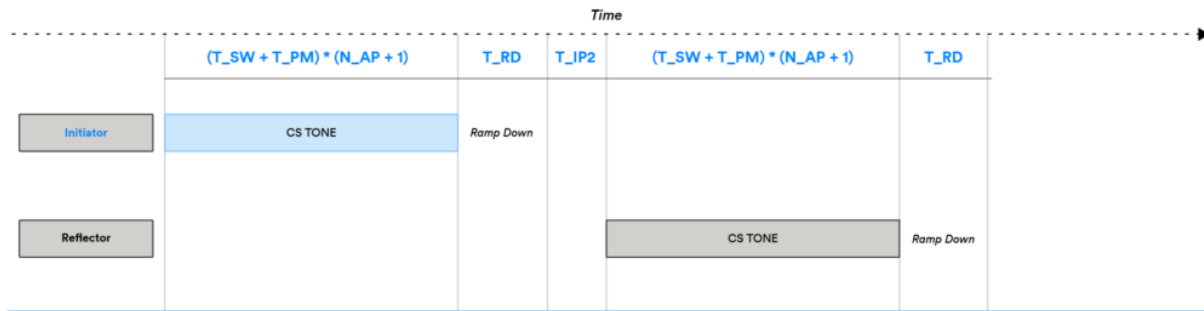


Figure 27 - mode-2 transmissions and time slots

The Initiator measures the phase of CS Tone received from the Reflector during the period T_{PM} , once for each antenna path. Adjustments are made using compensation values calculated in the mode-0 step. Phase measurements are passed to the application layer in an HCI event in the form of an array of IQ samples.

It should be noted that the expression for the total duration of the CS Tone(s) transmission includes the term $N_{AP} + 1$. This is because an extra time period known as a CS Tone extension slot follows the T_{PM} duration time slots that are allocated for each antenna path. Use of this time slot for transmission is randomized for security reasons, but, when it is used, a CS Tone is transmitted using the same antenna used in the immediately prior T_{PM} time slot.

Support for mode-2 steps is mandatory.

3.5.3.4 Mode-3

A mode-3 step supports both RTT calculation and PBR using combined exchanges of CS_Sync packets and CS Tones.

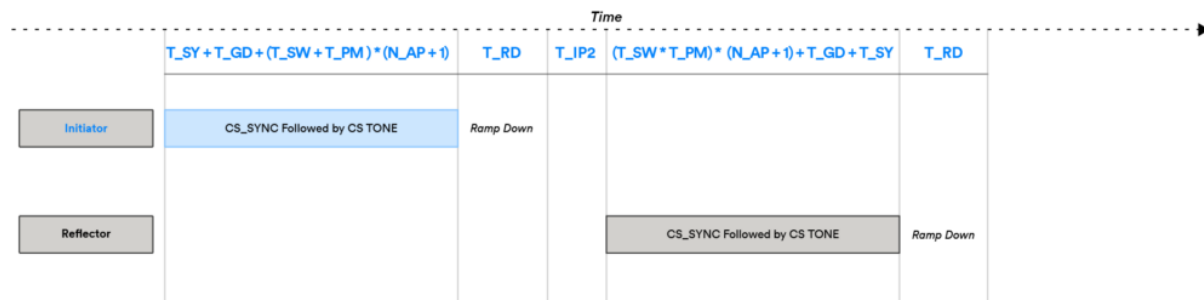


Figure 28 - mode-3 transmissions and time slots

Support for mode-3 is not mandatory. Applications wishing to combine PBR and RTT but finding through the capabilities exchange procedure that mode-3 is not supported by both Initiator and Reflector may instead use a mode sequence that combines both mode-2 and mode-1 steps.

Mode-3 steps include an extension slot as was described for mode-2 steps.

3.6 Establishing Phase Differences

In the previous sections on mode-0, mode-1, mode-2, and mode-3 steps, the focus was on the details of how time is divided and used in a single step of each type. But distance calculations require multiple exchanges, either to improve the accuracy of the calculated distance or because the method used demands it. PBR by definition, needs at least two exchanges.

For phase differences to be available for measurement, there needs to be more than one transmitted signal and more than one frequency must be involved. A single step involves the exchange of a single CS Tone on a single selected channel and frequency. As such, it is clear that the PBR method requires the execution of an absolute minimum of two steps of a mode that supports the PBR method. It should be noted that in general terms, a larger number of CS Tone exchanges using a correspondingly larger set of RF channels will provide the application with more data and the opportunity to produce more

accurate distance measurements. A larger number of exchanges will require more time to execute, however.

3.7 Antenna Switching

As noted in 3.2.3 Antenna Arrays, devices may include multiple antennas for use during phase-based ranging exchanges. The maximum number of antennas that a device may have for use during PBR exchanges (i.e., mode-2 or mode-3 steps) is four. A given pair of antenna configurations, one belonging to the Initiator and one to the Reflector, provides a number of antenna paths between the two devices.

A total of eight antenna permutations are defined by the Bluetooth Core Specification. Table 7 lists these configurations.

Antenna Configuration Index (ACI)	Device A number of antennas	Device B number of antennas	Number of antenna paths (N_AP)
0	1	1	1
1	2	1	2
2	3	1	3
3	4	1	4
4	1	2	2
5	1	3	3
6	1	4	4
7	2	2	4

Table 7 - Antenna Configurations

Antenna switching takes place during mode-2 steps (PBR) and during the PBR-related part of each mode-3 step. Specifically, it is when transmitting a CS Tone that antenna switching can be applied, depending on the antenna configuration of the transmitting device. The calculation of the duration of the CS Tone transmission time slots in mode-2 and mode-3 steps accommodates antenna switching and multiple antenna paths:

$$(T_{SW} + T_{PM}) * (N_{AP} + 1)$$

- T_{SW} provides time for antenna switching to take place and has a value of either 0, 2, 4, or 10 microseconds.
- T_{PM} is the time for the transmission of the CS Tone.
- N_{AP} is the number of antenna paths. The +1 term is to allow for the extension slot.

3.8 Mode Sequencing

3.8.1 Mode Sequencing Overview

A Bluetooth® Channel Sounding procedure always involves the execution of a sequence of multiple steps and a mix of at least two modes. The Bluetooth Core Specification defines mode combination and sequencing rules, key aspects of which shall be explored in this section.

Bluetooth® Channel Sounding applications will produce better quality, more accurate distance measurements when provided with data by the Bluetooth controller that is derived from larger numbers of exchanges of packets and tones.

3.8.2 Mode Combinations

Steps of at least two different mode types are always involved in a Bluetooth® Channel Sounding procedure. The first is the mode-0 step for frequency offset measurement and the second must be any one of the other modes. The primary non-mode-0 mode is called the Main_Mode. The secondary non-mode-0 mode, if there is one, is called the Sub_Mode. Table 8 lists the six permitted non-mode-0 mode combinations.

Main_Mode	Sub_Mode
Mode-1	None
Mode-2	None
Mode-3	None
Mode-2	Mode-1
Mode-2	Mode-3
Mode-3	Mode-2

Table 8 - Permitted non-mode-0 mode combinations

3.8.3 Mode Sequence Configuration and Sub_Mode Insertion

Applications are able to configure the step mode sequence using HCI commands. Amongst the key parameters that may be requested and agreed between devices are those shown in Table 9.

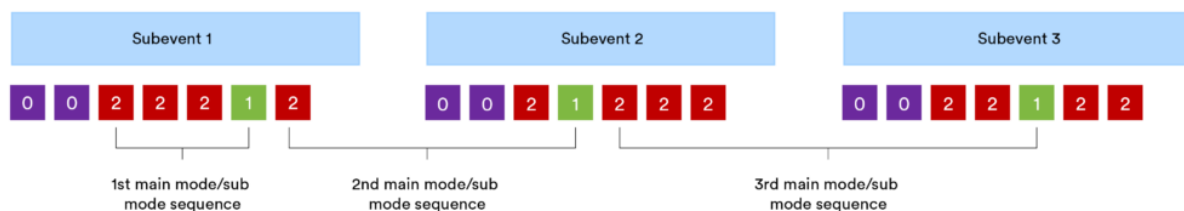
HCI Parameter	Purpose
Mode_0_Steps	Defines the number of consecutive mode-0 steps to be executed at the start of each CS subevent. Permitted values are 1, 2, or 3.
Main_Mode_Type	Indicates the mode which will be the main mode (1, 2, or 3).
Sub_Mode_Type	Indicates the mode which will be the submode (1, 2, or 3).
Min_Main_Mode_Steps	Determines the minimum number of main mode steps that must always be executed before a submode step.
Max_Main_Mode_Steps	Determines the maximum number of main mode steps that must always be executed before a submode step.

Table 9 - Mode sequencing control parameters

In general, step mode sequencing follows this pattern:

- 1 One or more mode-0 steps start the subevent
- 2 A sequence of n main mode steps then follows, where n is randomly selected and falls in the range of Min_Main_Mode_Steps to Max_Main_Mode_Steps inclusive
- 3 A single submode step follows the sequence of n main mode steps due to a process that the Bluetooth Core Specification calls sub_mode insertion

Step mode sequences are not tied to subevent boundaries other than by the general rule that subevents must always start with one or more mode-0 steps. Full sequences can span more than one subevent.



Mode Sequencing Parameters

Mode_0_Steps = 2, Main_Mode_Type = 2, Sub_Mode_Type = 1

Min_Main_Mode_Steps = 2, Max_Main_Mode_Steps = 5

Figure 33 - A step mode sequence example

3.8.4 Main Mode Repetition

There is another mode sequencing parameter that can be used by applications. Main_Mode_Repetition specifies a number of the most recent main mode steps from the last subevent to be repeated in the current subevent.

When main mode repetition applies, the steps repeated in the current subevent use the same channel index as was used for the respective steps in the previous subevent. This ensures that the repeated step transmissions have the same intended frequency. Note that the purpose of repeating main mode steps on the same frequency is to address possible frequency drift as well as Doppler shift effects.

Main mode repetition affords applications the opportunity to correlate some of the properties of the exchanges and may make it easier to track the velocity of moving devices.

Steps that are incorporated into the mode sequence due to main mode repetition are not counted in the submode insertion process.

3.8.5 Applications and Mode Sequencing Considerations

The ability to configure mode combinations and control step mode sequences using submode insertion and main mode repetition gives applications a lot of control over the process of Bluetooth® Channel Sounding. PBR is the most accurate of the two distance measurement methods and using RTT at the same time adds considerable security to the system. It also allows distance ambiguity which can arise using the PBR method to be dealt with.

Step mode-3 provides support for both methods in a single mode type but support for mode-3 is optional. Therefore devices that discover during the capabilities exchange procedure that mode-3 is not available, must mix mode-1 (RTT) and mode-2 (PBR) steps. This can be achieved by selecting mode-2 as the main mode and mode-1 as the submode.

A further consideration for applications is latency. Each exchange of signals takes time. Mode-1 RTT exchanges sometimes take longer than Mode-2 PBR exchanges depending on the number of antenna paths. Applications that need to keep latency below a certain threshold are likely to opt to include a lower proportion of RTT exchanges in the Bluetooth® Channel Sounding procedure.

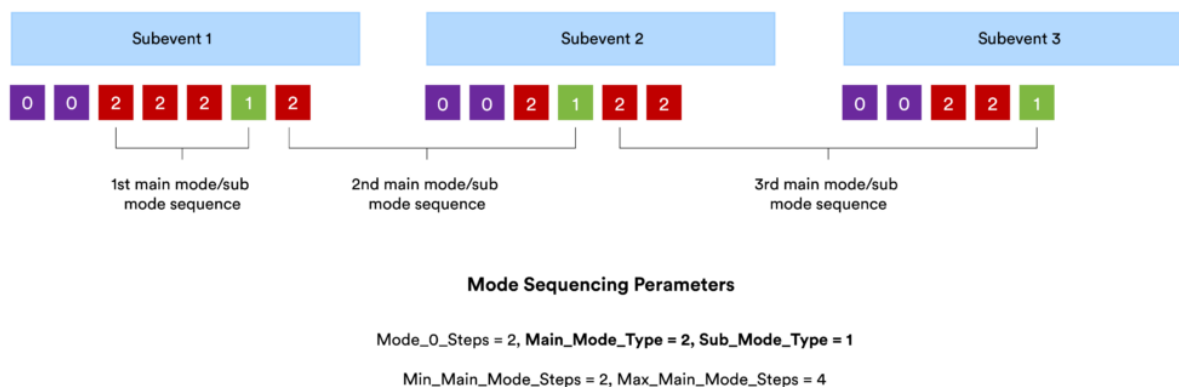


Figure 34 - A 3:1 PBR to RTT ratio using mode-2 and mode-1

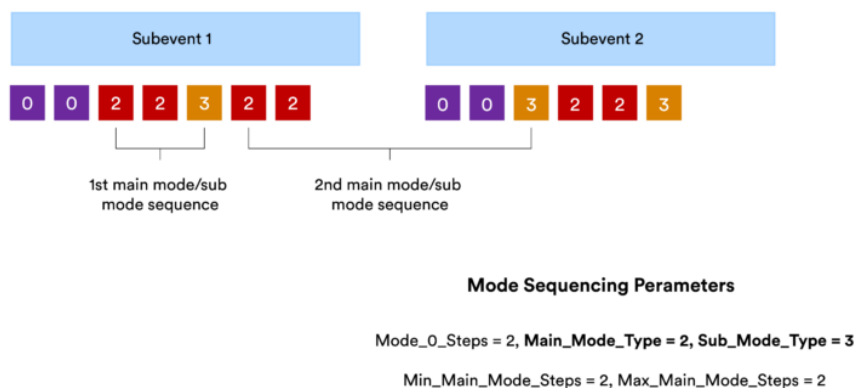


Figure 35 - A 3:1 PBR to RTT ratio using mode-2 and mode-3

3.9 RF Channels and Channel Selection

3.9.1 The Bluetooth® Channel Sounding Channel Map

Typically, Bluetooth LE divides the 2.4 GHz ISM band into 40 channels, each 2 MHz wide. This is not the case when using Bluetooth® Channel Sounding, however.

For the purposes of Bluetooth® Channel Sounding, 72 channels are defined, each with a 1 MHz width and a unique channel index value. The arrangement of these channels ensures that the LE primary advertising channels are avoided.

A channel width of 1 MHz rather than the usual 2 MHz ensures that the frequency separation between PBR signals that use adjacent channels is such that distance ambiguity does not arise until around 150 meters.

CS Channel Index	RF Center Frequency	Allowed
0	2402 MHz	No
1	2403 MHz	No
2	2404 MHz	Yes
...	...	...
22	2424 MHz	Yes
23	2425 MHz	No
24	2426 MHz	No
25	2427 MHz	No
26	2428 MHz	Yes
...	...	...
76	2478 MHz	Yes
77	2479 MHz	No
78	2480 MHz	No

Table 10 - Bluetooth® Channel Sounding channel indices and RF physical channels

3.9.2 Channel Filtering

A channel index filter bit map is maintained. This is a list of the channel indices defined for Bluetooth® Channel Sounding with each channel marked as either included or excluded. The Bluetooth® Channel Sounding channel index filter map is maintained by a link layer procedure called Channel Sounding Channel Map Update procedure which allows either the Initiator or Reflector to inform the other device about which channels to use or avoid, based on its assessment of local channel conditions. An excluded channel is never selected by any of the channel selection algorithms.

3.9.3 Frequency Hopping

Frequency hopping generally takes place immediately before the execution of a step, as depicted in Figure 36.

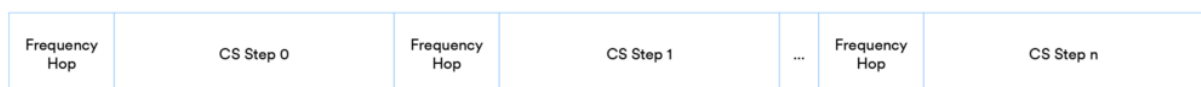


Figure 36 - Frequency hopping ahead of step execution

An exception to this rule applies when mode repetition has been configured with a non-zero value assigned to the Main_Mode_Repetition parameter. Steps repeated due to mode repetition will use the same channel index as the step in the previous subevent that they repeat.

3.9.4 Channel Selection

3.9.4.1 Overview

A new set of three channel selection algorithms (CSA) has been defined for use in Bluetooth® Channel Sounding. Collectively they are known as CSA #3 and individually as CSA #3a, CSA #3b, and CSA #3c.

CSA #3a is used solely for selecting the channel to use in mode-0 steps. CSA #3b and CSA #3c are both designed for use with non-mode-0 steps but only one of the two may be used during a Bluetooth® Channel Sounding procedure instance. Consequently, two different channel selection algorithms are actively associated with Bluetooth® Channel Sounding at any time.

3.9.4.2 Channel Index Shuffling

Channel selection involves two distinct channel index lists. The first is used by CSA #3a and the selection of channels for mode-0 steps. The second is used for non-mode-0 steps with CSA #3b or CSA #3c.

Channel index lists are created by randomizing the order of those channels marked as included in the channel map to create a shuffled channel list. CSA #3a and CSA #3b do this in exactly the same way. CSA #3c takes a different approach but relies on the same primitive shuffling function, known as cr1 in the Bluetooth Core Specification.

3.9.4.3 CSA #3a

The mode-0 channel selection algorithm CSA #3a uses a shuffled channel list. The shuffled channel list used for mode-0 step frequency hopping is distinct from the corresponding list of channels used for non-mode-0 channel hopping. Each entry in the shuffled channel list is unique and used only once. When all entries in the shuffled channel list have been used it is regenerated, creating a new randomized list of channels.

3.9.4.4 CSA #3b

The non-mode-0 channel selection algorithm CSA #3b uses a shuffled channel list that is distinct from the corresponding list of channels used for mode-0 channel hopping. CSA #3b allows the channel index list to be iterated more than once before it is regenerated, and this is controlled by a parameter called CSNumRepetitions which applications may set.

3.9.4.5 CSA #3c

Algorithm CSA #3c is significantly different to CSA #3b. Subsets of the included channels in the channel map are organized into groups and channel patterns generated which form shapes. Two pattern types are supported and named hat and X. CSA #3c may offer some advantages in detecting reflected signal paths in some circumstances. Support for CSA #3c is optional.

3.10 RTT Options and Accuracy

The RTT method involves the exchange of CS_Sync packets in steps of mode-1 and/or mode-3. Several ways of establishing the Time of Arrival (ToA) timestamps, that are needed for the calculation of round-trip times, are defined. An application can indicate the method to be used during the Bluetooth® Channel Sounding configuration procedure via HCI commands using the RTT_Type parameter.

The options are to base timing measurements on the Access Address field, to use a sounding sequence that is either 32 or 96 bits in length, or to use a random sequence which may be 32, 64, 96, or 128 bits in length. The accuracy of time estimates varies according to the method used and the length of the field used for timing purposes. Both the use of a sounding sequence and the use of a random sequence allow a more accurate form of estimate known as a fractional timing estimate to be made.

3.10.1 Timing Based on an Access Address

CS_Sync packets contain a 32-bit Access Address field. The simplest method which may be used to establish a ToA value involves the controller using its clock to capture a timestamp at the time when the Access Address field in a CS_Sync packet has been received.

An Access Address is a 32-bit binary value at the link layer but when transmitted its value is represented by a series of analog symbols, formed by applying GFSK modulation to those digital bits. A single symbol consists of a radio transmission at a frequency that represents a 0- or 1-bit value and that depending on the symbol rate, has a duration of either one microsecond or half a microsecond.

The transmitter's oscillator and the receiver's oscillator are unlikely to be in phase with each other and this can be a source of inaccuracies in this process. To improve results, it is suggested that measurements are taken over a series of packets exchanged in a sequence of steps and the distribution of values calculated. This distribution can then be used to improve the accuracy of ToA timestamps.

3.10.2 Fractional Timing Estimates

Two optional methods which offer better ToA timestamp accuracy are described in the Link Layer Specification, Part H, sections 3.3 and 3.4. Both provide fractional timing estimates.

CS_Sync packets can accommodate extra, optional data at the end of the packet. If this option is used, one of two fields may be appended to the CS_Sync packet: either a Random Sequence or a Sounding Sequence.

The first of the fractional timing methods involves analyzing the optional Random Sequence field in CS_Sync packets to determine fractional timing errors. The fractional timing error calculated from the Random Sequence is used to optimize the Access Address timestamp.

The second fractional timing method is based on an analysis of a Sounding Sequence field appended to a CS_Sync packet. A sounding sequence is an alternating pattern of 0s and 1s at the link layer which when modulated using GFSK produces two distinct radio tones with different frequencies and differing phases. An analysis of the phase difference exhibited by the two tones allows a fractional timing error to be calculated and used to optimize the ToA timestamp.

3.10.3 A Comparison of RTT Methods

Bluetooth® Channel Sounding application developers can derive round-trip times based on measurements produced by one of three different methods. The choices are to use the ToA of an Access Address or to use one of two fractional methods based on either a random sequence or a sounding sequence in a CS_Sync packet.

The three RTT methods offer differing degrees of distance measurement accuracy, security and latency for application developers. In general, the fractional methods have the potential to deliver the most accurate results and the best security.

3.11 The LE 2M 2BT PHY

3.11.1 Modulation Schemes

A modulation scheme defines a means of encoding digital information in signals using one or more of the signal's physical properties. Frequency Shift Keying (FSK) is a simple example of a modulation scheme. It involves producing a symbol that represents a binary value of 1 by shifting the frequency of a carrier signal up by a certain amount (the frequency deviation) or down by the same amount to represent a binary 0.

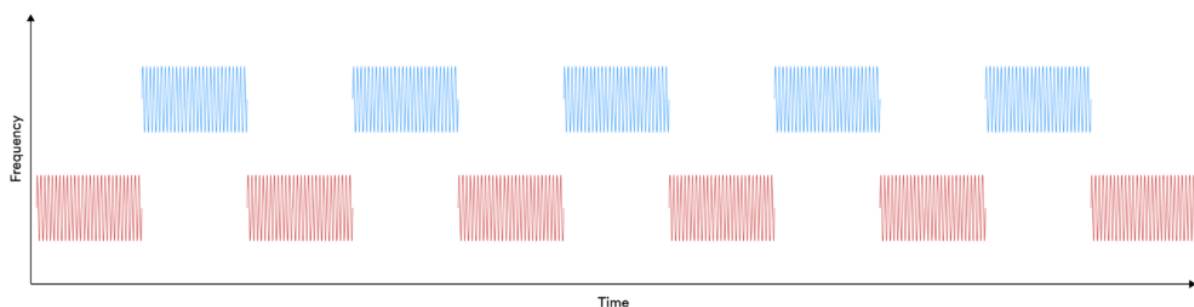


Figure 37 - Frequency Shift Keying (FSK) encoded bit stream 01010101010

The abrupt switching between frequencies that is a feature of basic FSK results in the generation of noise that spreads over a wider range of frequencies than is desirable. To counter this, Bluetooth technology uses a special variant of FSK called Gaussian Frequency Shift Keying (GFSK). GFSK differs from basic FSK in that it involves a filter that causes the transition between frequencies to follow a curve. The shape of the curve and the rate of frequency transitions is determined by various parameters, including the Bandwidth-Bit Period Product or BT.

3.11.2 Bandwidth-Bit Period Product

The Bandwidth-Bit Period Product (BT) is an attribute of a signal that provides information about the relationship between its bandwidth and the duration of symbols. BT affects the shape and span of the radio pulses that constitute symbols. A higher BT value results in a narrower, squarer pulse and a lower value results in a wider, more rounded pulse shape.

3.11.3 LE 2M 2BT

The Bluetooth Core Specification introduces a new PHY called LE 2M 2BT. LE 2M 2BT may currently only be used with Bluetooth® Channel Sounding. A comparison of PHYs with key aspects of LE 2M 2BT highlighted appears in Table 11.

	LE 1M	LE Coded	LE 2M	LE 2M 2BT
Symbol Rate	1 Msym/s	1 Msym/s	2 Msym/s	2 Msym/s
BT	0.5	0.5	0.5	2.0
Min. Frequency Deviation	185 kHz	185 kHz	370 kHz	420 kHz
Error Detection	CRC	CRC	CRC	N/A
Error Correction	NONE	FEC	NONE	N/A
Requirement	Mandatory	Optional	Optional	Optional. Only to be used with Channel Sounding.

Table 11 - A comparison of Bluetooth LE PHYs

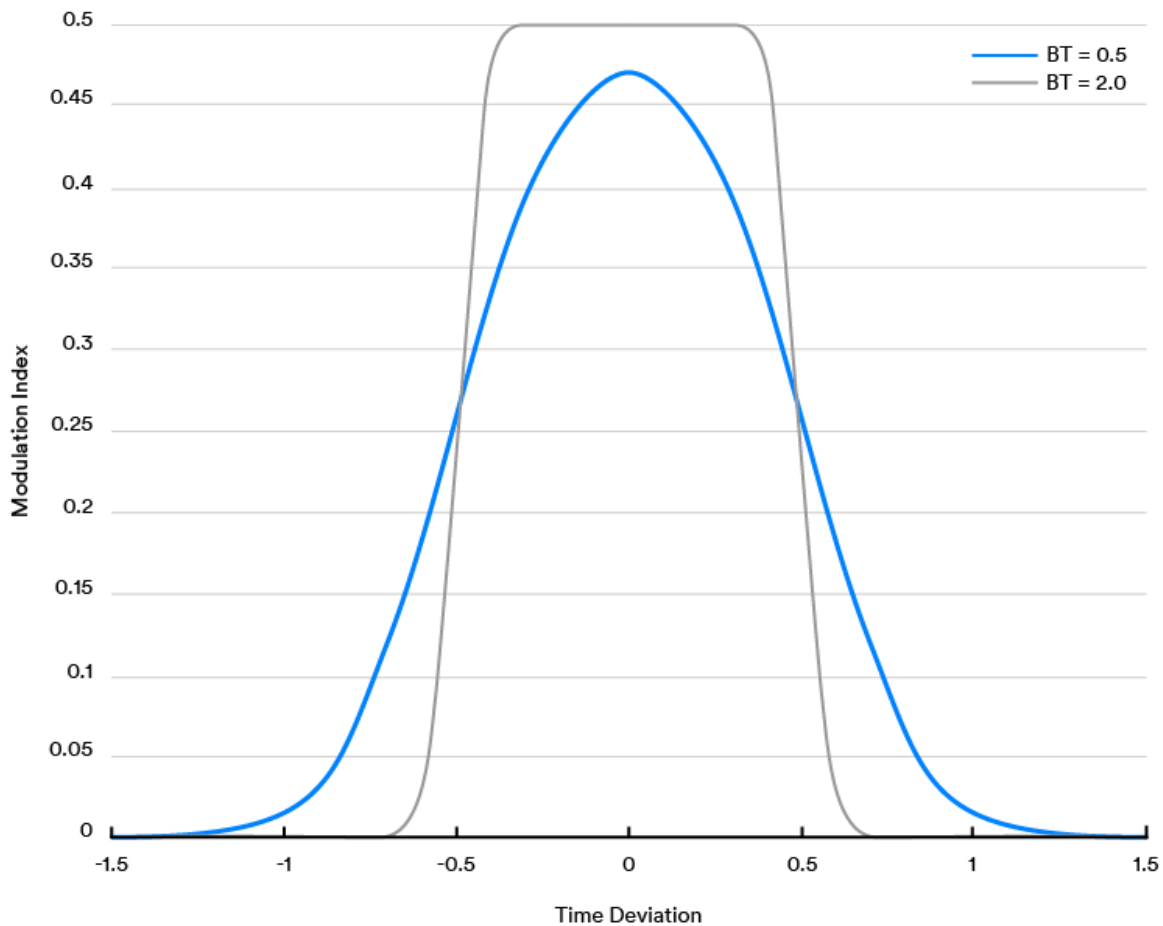


Figure 38 - Pulse Shapes

3.12 SNR Control for Bluetooth® Channel Sounding Steps

Some radio transmitters have the ability to adjust their signal-to-noise ratio (SNR) to lie within a specified range. This capability, if supported by both the Initiator and Reflector devices, can be used to improve the security of Bluetooth® Channel Sounding steps that relate to the RTT distance measurement method i.e. mode-1 and mode-3 steps.

The Bluetooth Core Specification defines a number of SNR Output Index (SOI) values that correspond to associated SNR output levels measured in dB. Table 12 reproduces these definitions.

SNR Output Index (SOI)	SNR Output Level (dB)
0	18
1	21
2	24
3	27
4	30

Table 12 - SNR Output Indices and Levels

3.13 Security

3.13.1 Overview

Security issues that are unique to distance measurement solutions generally involve the threat of untrusted devices somehow tricking one trusted device into concluding that another trusted device is sufficiently close for some action to be granted or taken.

Bluetooth® Channel Sounding includes a collection of features which can act as countermeasures to a number of distance measurement security threats. These features can be regarded as falling into four categories:

- 1 Using PBR and RTT methods in combination
- 2 Randomization of the bit stream and transmission patterns
- 3 Defense against symbol manipulation
- 4 RF signal analysis techniques including attack detection

3.13.2 PBR and RTT Cross Checking

Bluetooth® Channel Sounding supports two distance measurement methods, phase-based ranging (PBR) and round-trip timing (RTT). The two methods work completely differently. An application may use both methods in conjunction by selecting a suitable mode combination such as mode-2 as the main mode for PBR and mode-1 as the submode for RTT.

The complexity of attacking both methods simultaneously so that both the phase of the Bluetooth® Channel Sounding signals and calculated round-trip times are manipulated to give misleading and consistent results is regarded by security experts as very high.

3.13.3 Initializing Bluetooth® Channel Sounding Security

Firstly, devices must have been paired with each other. This is necessary for it to be possible for an encrypted LE-ACL link to be created. CS Security Start then takes place over the encrypted LE-ACL link which means that the exchange of Bluetooth® Channel Sounding security key data is protected from eavesdroppers.

Finally, both the Central and Peripheral devices perform a secure exchange of partial values of Bluetooth® Channel Sounding security data. This provides both devices with the same data from which to construct a complete and common value for each of the CS initialization vector (CS_IV), CS instantiation nonce (CS_IN) and CS personalization vector (CS_PV).

3.13.4 Deterministic Random Bit Generator (DRBG)

The Bluetooth Core Specification defines a random bit generator that is consistent with the recommendations defined in NIST Special Publication 800-90Ar1. It is known as the Deterministic Random Bit Generator or DRBG.

Instantiating DRBG requires the three Bluetooth® Channel Sounding security parameters, CS_IV, CS_IN and CS_PV to be provided as inputs. Having executed the Bluetooth® Channel Sounding Security Start procedure, both the Initiator and Reflector device possess the same values for these parameters. When initialized with the same parameters values, two instances of DRBG will produce exactly the same bit sequences over a series of invocations and it is this which makes the algorithm deterministic.

3.13.4.1 Secure Access Addresses

In the case of Bluetooth® Channel Sounding, each device changes its Access Address field in CS_Sync packets at every mode-0, mode-1 and mode-3 CS step. New Access Address values are generated using selection rules that involve the DRBG and both devices know the Access Address that will be used by the other. Receiving devices check the Access Address value and report any issues to the host.

The Access Address field is 32 bits in length and can have 4,294,967,296 different values. A malicious device wishing to spoof a CS_Sync packet will therefore have a 1 in 4,294,967,296 chance of guessing the correct Access Address value in each one of the multiple CS_Sync packets exchanged.

3.13.4.2 Random Sequence for RTT Fractional Timing

CS_Sync packets can include an optional Random Sequence field. This field supports one of the fractional RTT methods. The content of the Random Sequence field is regenerated using the CS DRBG for every transmitted CS_Sync packet. The Random Sequence field may be 32, 64, 96, or 128 bits in length.

3.13.4.3 Sounding Sequence Marker Signals

A Sounding Sequence consists of a predictable alternating pattern of 32 or 96-bits and is used for fractional RTT calculations. To mitigate the risk of this known bit pattern being exploited, DRBG is used to select positions within the sequence to insert one of two randomly selected 4-bit values known as marker signals. Marker signals selected by DRBG have a value of either 0b1100 or 0b0011. The random insertion of random bit patterns within a sounding sequence protects against sounding sequence spoofing.

3.13.4.4 Tone Extension Slot Random Transmissions

Mode-2 and mode-3 steps include a tone extension slot. The tone extension slot is always reserved but whether or not a transmission takes place in that time slot is randomized and governed by DRBG. The receiving device knows when to expect and when not to expect a transmission in the tone extension slot but an attacking device does not.

3.13.4.5 Random Selection of Antenna Paths

During phase-based ranging, a tone is transmitted over every available antenna path that exists between the two devices. The sequence of paths used is randomized using DRBG at every Bluetooth® Channel Sounding step.

3.13.5 Sounding Sequences

A sounding sequence consists of a sequence of alternating bit values of 0 and 1. The corresponding RF signal can be viewed as consisting of two tones of different frequencies and with differing phases. PBR calculations can therefore be made using the phase differences of the two tones encoded in the Sounding Sequence field of a single CS_Sync packet simultaneously while using the CS_Sync packet to calculate a round-trip time. The simultaneous calculation of both RTT and PBR measurements based on a single packet makes an attempt to attack the exchange extremely complex.

3.13.6 Attack Detection and Reporting

The Bluetooth® Channel Sounding section of the Link Layer specification includes a description of an attack detector system. Bluetooth® Channel Sounding attack detection in the Bluetooth controller is based on the evaluation of received signals against a reference signal definition and the examination of the received signal for indicators of a possible attack such as unexpected bit transitions or phase adjustments.

A standardized metric for reporting the probability of an attack being underway is defined by the Bluetooth Core Specification and is called the Normalized Attack Detector Metric or NADM. Table 13 contains the NADM value definitions.

NADM Value	Description
0x00	Attack is extremely unlikely
0x01	Attack is very unlikely
0x02	Attack is unlikely
0x03	Attack is possible
0x04	Attack is likely
0x05	Attack is very likely
0x06	Attack is extremely likely
0xFF	Unknown NADM. Default value for RTT types that do not have a random sequence or sounding sequence.

Table 13 - NADM values

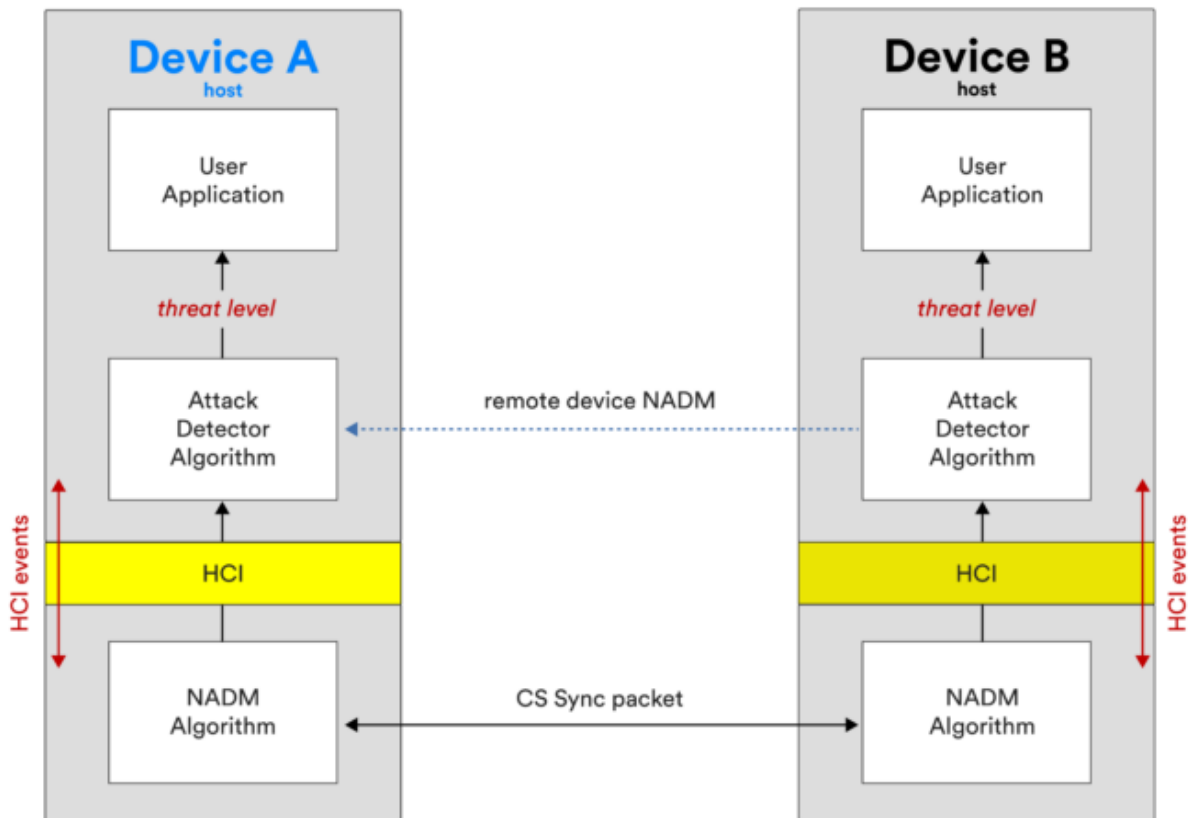


Figure 39 - Attack Detector System Outline

3.13.7 LE 2M 2BT

There exist a number of known physical layer attacks that involve a man-in-the-middle (MITM) attacker anticipating the value of partially received symbols from a legitimate transmitting device and relaying full, generated versions of those symbols with the timing manipulated so that the legitimate recipient miscalculates the round-trip time. The LE 2M 2BT PHY with its bandwidth bit-period product value of 2.0 involves symbol pulses that have a duration that is shorter than the pulses associated with the other PHYs and this lessens the risk of these types of attack.

3.13.8 SNR Control and RTT Security

The SNR Control feature allows the Initiator and Receiver to mix a pre-agreed amount of random noise into signals. This applies only to CS_Sync packet transmissions made during mode-1 (RTT) and mode-3 (RTT and PBR) steps. By injecting noise into the signal, it becomes harder and slower for the attacker's analysis to complete and thus reduces the likelihood of such attacks succeeding. On the other hand, the Initiator and Reflector devices, having pre-agreed the SNR, are able to filter the artificially added noise easily.

3.13.9 CS Security Levels

The Generic Access Profile (GAP) section of the Bluetooth Core Specification defines security modes and security levels. A formal definition of four security levels for Bluetooth® Channel Sounding is included. It is likely that future Bluetooth profile specifications will reference these definitions.

3.13.10 Vendor-Specific Implementations and Additional Security

Controller implementers may opt to introduce further vendor-specific security measures.

3.13.11 Channel Sounding Amplitude-based Attack Resilience

Amplitude-based attack resilience strengthens the security capabilities of Bluetooth® Channel Sounding. By defining a precise attack model, a robust DFT-based detection metric, and a thorough multi-stage testing regimen, it ensures that certified devices can reliably detect sophisticated attempts to manipulate distance measurements.

Coupled with necessary updates to the HCI and Link Layer protocols, the Channel Sounding Amplitude-based Attack Resilience feature provides a comprehensive framework for mitigating evolving threats. It reflects Bluetooth's commitment to security, helping to ensure that its precision ranging technology remains both accurate and trustworthy in the face of new challenges.

3.14 Host Applications

Creating Bluetooth fine-ranging applications and products involves harnessing the Bluetooth® Channel Sounding feature of the controller and combining it with custom, application-layer code. Developers of the application component of a solution must take care of various issues which this section highlights.

3.14.1 The Distance Measurement Algorithm

The Bluetooth stack does not generate distance measurements directly. Instead, during the execution of CS steps by the Bluetooth controller, low-level measurements of phase and/or timing are made and it is from this data that applications can calculate distance measurements.

The algorithm which applications use for calculating distances is not specified by the Bluetooth Core Specification. Consequently, this is one area in which vendors can differentiate. Superior algorithms will produce superior results.

3.14.2 Controller to Host Communication of Bluetooth® Channel Sounding Data

3.14.2.1 HCI Event Types

The Host Controller Interface Functional Specification defines two events that are used by a controller to pass Bluetooth® Channel Sounding data to the host. The two events are called LE CS Subevent Result and LE CS Subevent Result Continue.

3.14.2.2 HCI Event Timing

The controller aggregates the measurements generated during the steps executed within a Bluetooth® Channel Sounding subevent. Complete or partial sets of results are reported using a LE CS Subevent Result HCI event. If an incomplete set is reported, the remainder of the results are reported in one or more LE CS Subevent Result Continue events that are sent later. The HCI event fields `Subevent_Done_Status` and `Procedure_Done_Status` indicate to the application layer whether all data for the subevent or procedure has been reported or whether there is more to come.

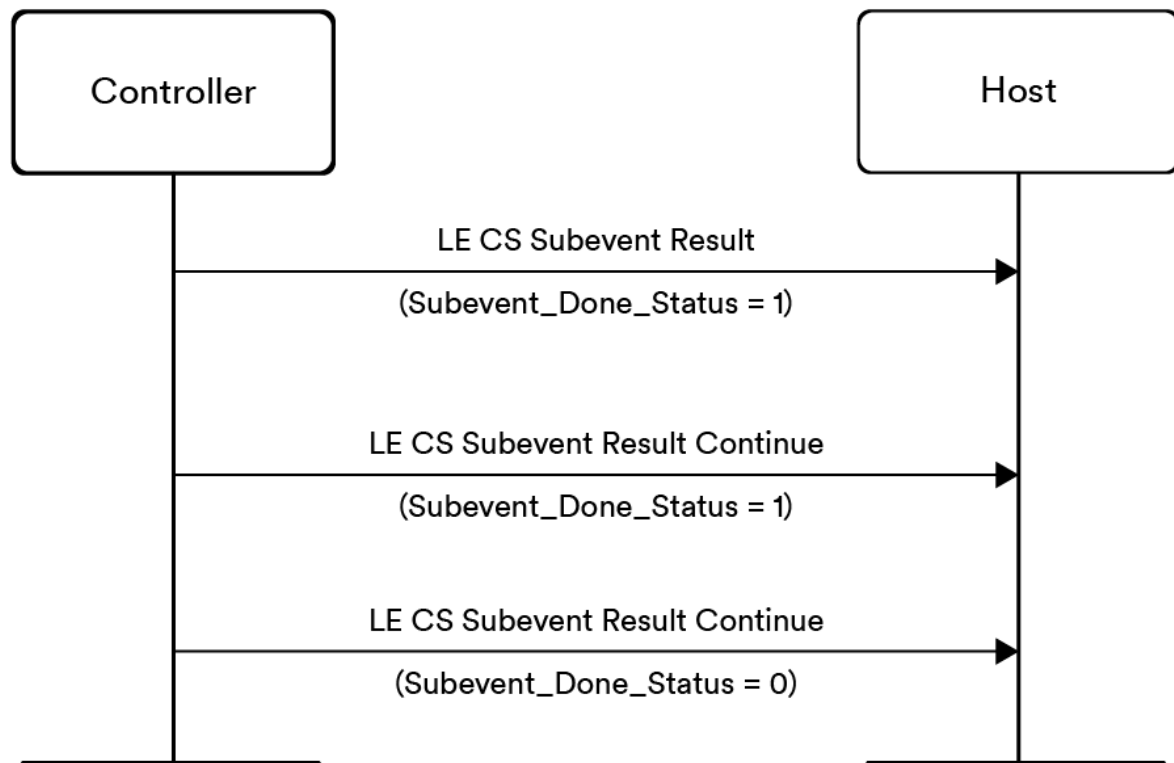


Figure 40 - Example Bluetooth® Channel Sounding HCI data reporting

3.14.2.3 HCI Event Content

The Bluetooth® Channel Sounding HCI events convey a range of types of data from the controller to the host. A selection of key fields and data structures are described here.

Frequency_Compensation - The purpose of mode-0 steps is to determine differences between wanted and actual frequencies generated by the Initiator and Reflector. This is used to calculate a fractional frequency offset (FFO) which can then be used to compensate for the impact that such differences have both on frequency and timing values and ultimately to improve the accuracy of distance measurements.

Num_Steps_Reported - This field indicates how many steps are being reported on in this HCI event. It also indicates the size of four step related arrays of data: Step_Mode, Step_Channel, Step_Data_Length, and Step_Data.

Step_Mode [] - This array contains the mode of each step, ordered by step number and expressed as a value in the range 0 - 3.

Step_Channel [] - This array contains the index of the RF channel used in the execution of the corresponding step.

Step_Data_Length [] - The data reported for each step is variable in terms of its content and structure. This array contains the length of each element in the associated Step_Data array.

Step_Data [] - The data reported for each step depends on the step mode, the device role (Initiator or Reflector) and whether or not a sounding sequence is used. The structure containing the relevant data is called the Mode_Role_Specific_Info object and eleven variants of this structure are defined. Examples of data which may be found include Packet_Quality, Tone_Quality, the received signal strength indicator (RSSI), the measured frequency offset, a NADM value, the antenna identifier, phase correction terms and elapsed time measurements between packet transmission and arrival. Time values are expressed as a multiple of units of half a nanosecond.

3.14.3 Mode Combinations and Mode Sequencing

The application layer is responsible for deciding which step modes to use and where both a primary mode and a submode are used, what the ratio between the number of steps of each of the selected modes should be. Application or product developers will need to consider distance measurement accuracy requirements, security and latency in arriving at conclusions as well as the features supported by the local controller.

3.14.4 Application Layer Security

The application layer can exercise some control over the security of the overall solution in its selection of mode combinations and RTT parameters. Developers should first seek to understand and evaluate the security levels defined by the Generic Access Profile (GAP) as a start-point for establishing what security options to adopt.

It is recommended that PBR and RTT are always used in combination so that cross-checking of distance calculations based on the two methods can be used. PBR is supported by Bluetooth® Channel Sounding to offer the most accurate distance measurements whilst the primary reason for also supporting RTT is as a security measure.

NADM values are created by the NADM algorithm in the Bluetooth controller and a standardized adjective form of meanings is defined for these values. But it is the application layer which must decide what action to take (if any) for each of the possible NADM values.

3.15 Bluetooth Channel Sounding enhancements

Bluetooth® Channel Sounding is designed as an extensible framework that continues to evolve through incremental feature enhancements across Core Specification updates. While the core architecture and procedures establish a common foundation, additional capabilities may be introduced over time to address newly identified requirements, operational considerations, or implementation refinements. This section summarizes Bluetooth® Channel Sounding enhancements that extend specific aspects of the version 1 features found in Core Specification v6.0. As Channel Sounding continues to evolve, this section is intended to expand to incorporate future enhancements.

3.15.1 Resilience Against Amplitude-Based Attacks

Bluetooth Core Specification version 6.2 introduces enhanced resilience against a class of ranging attacks that exploit amplitude manipulation to influence distance measurements in Channel Sounding. These attacks apply a periodic amplification pattern synchronized to the symbol timing of Channel Sounding packets, causing predictable phase distortions in the receiver through amplitude-to-phase conversion in the analog front end. To detect such behavior, the specification extends the existing Normalized Attack Detector Metric (NADM) framework with a Discrete Fourier Transform (DFT)-based detection method. This approach analyzes energy components at the symbol frequency and its harmonics, which are characteristic of periodic amplitude manipulation, and incorporates the resulting metric into existing NADM reporting. The enhancement strengthens Channel Sounding's ability to detect advanced early-commit ranging attacks without prescribing specific receiver implementations, maintaining flexibility while addressing newly identified threat vectors.

3.15.2 Inline PCT Transfer Support

Bluetooth Core Specification version 6.3 introduces support for an inline Phase Correction Term (inline PCT) as an enhancement to the Channel Sounding phase-based ranging procedure. Inline PCT allows a reflector device to apply a phase correction term within the radio processing chain prior to retransmission, enabling phase-aligned signal forwarding without requiring all phase compensation to be performed digitally at the initiator. By incorporating phase correction inline, this mechanism can reduce control and reporting overhead associated with phase error compensation while preserving the fundamental two-way Channel Sounding measurement model. Support for inline PCT is optional and negotiated between devices, maintaining interoperability with implementations that rely on initiator-side digital phase correction. To learn more, refer to [section 3 of the Bluetooth® Core Specification v6.3 technical overview](#).

3.15.3 PHY-Specific RTT Accuracy Support

Bluetooth Core Specification version 6.3 enhances Channel Sounding Round-Trip Time (RTT) measurements by introducing PHY-specific RTT accuracy declarations. Prior to this enhancement, devices reported a single RTT accuracy target that applied uniformly across all supported PHYs, despite differing symbol rates and timing characteristics. The updated mechanism allows devices to declare RTT accuracy requirements independently for each PHY, including LE 1M and optional higher-rate PHYs such as LE 2M. This enables more accurate alignment between RTT exchange counts and the achievable timing precision of each PHY, improving efficiency and consistency in multi-PHY environments. The enhancement is supported through updated Link Layer capability exchanges and versioned Host Controller Interface commands, maintaining interoperability with legacy implementations. To learn more, refer to [section 3 of the Bluetooth® Core Specification v6.3 technical overview](#).

4. A Summary of Bluetooth Core Specification Changes

To introduce the Bluetooth® Channel Sounding feature, changes have been made to several layers of the Bluetooth Core Specification. A summary of key changes is presented in this section with the intention that this provide a chapter-by-chapter high-level reference for orientation purposes only. The Bluetooth Core Specification should be consulted for full details.

4.1 Architecture

Volume 1, Part A of the Bluetooth Core Specification describes the architecture of the technology.

- Section 3, Transport Architecture introduces a new packet structure and signaling format for Bluetooth® Channel Sounding. It also defines the new LE Channel Sounding Physical Chan[Truncated]